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OR

CBSE 2023

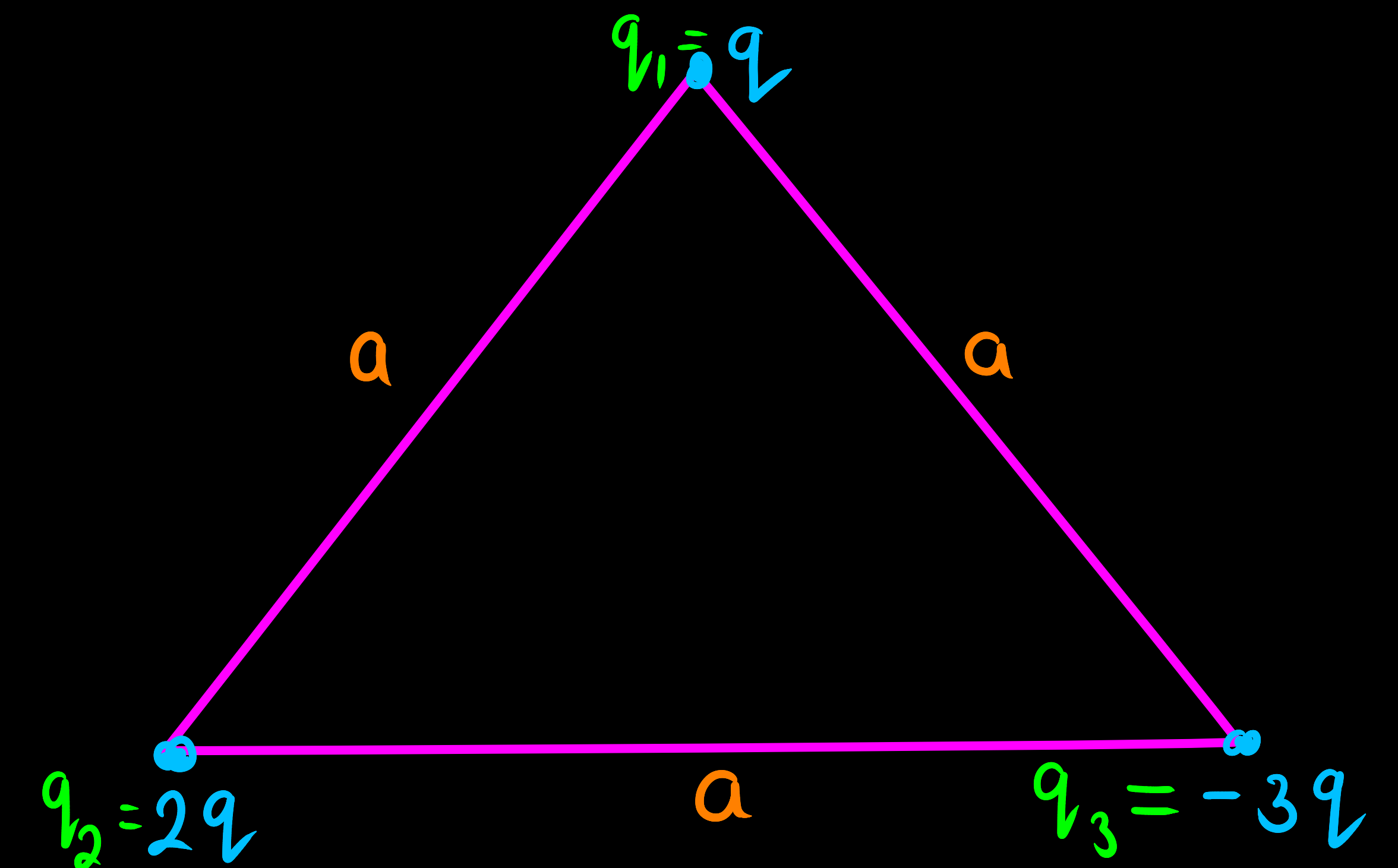
- (b) Two small conducting balls A and B of radius r_1 and r_2 have charges q_1 and q_2 respectively. They are connected by a wire. Obtain the expression for charges on A and B, in equilibrium. 2

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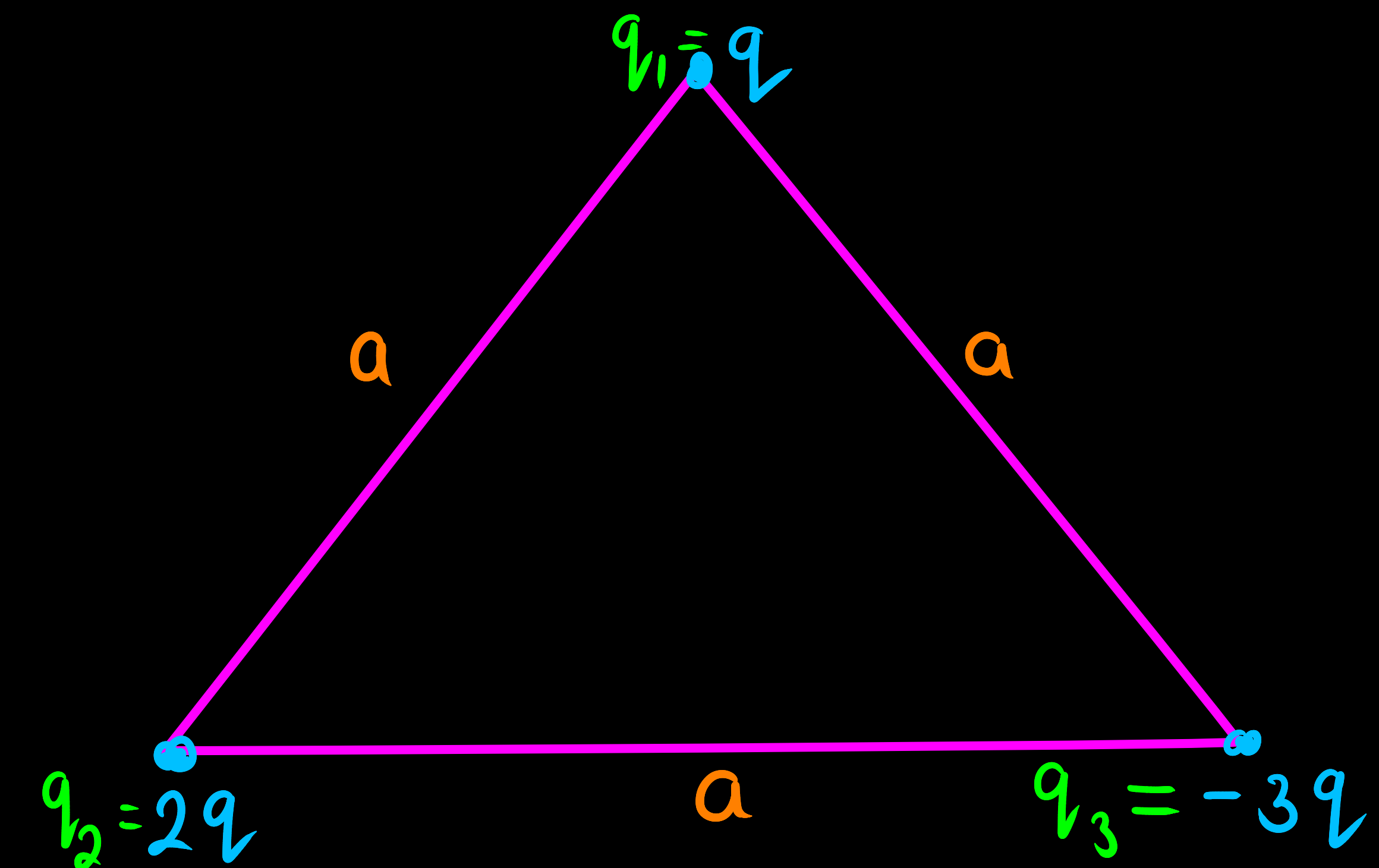
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CBSE 2023

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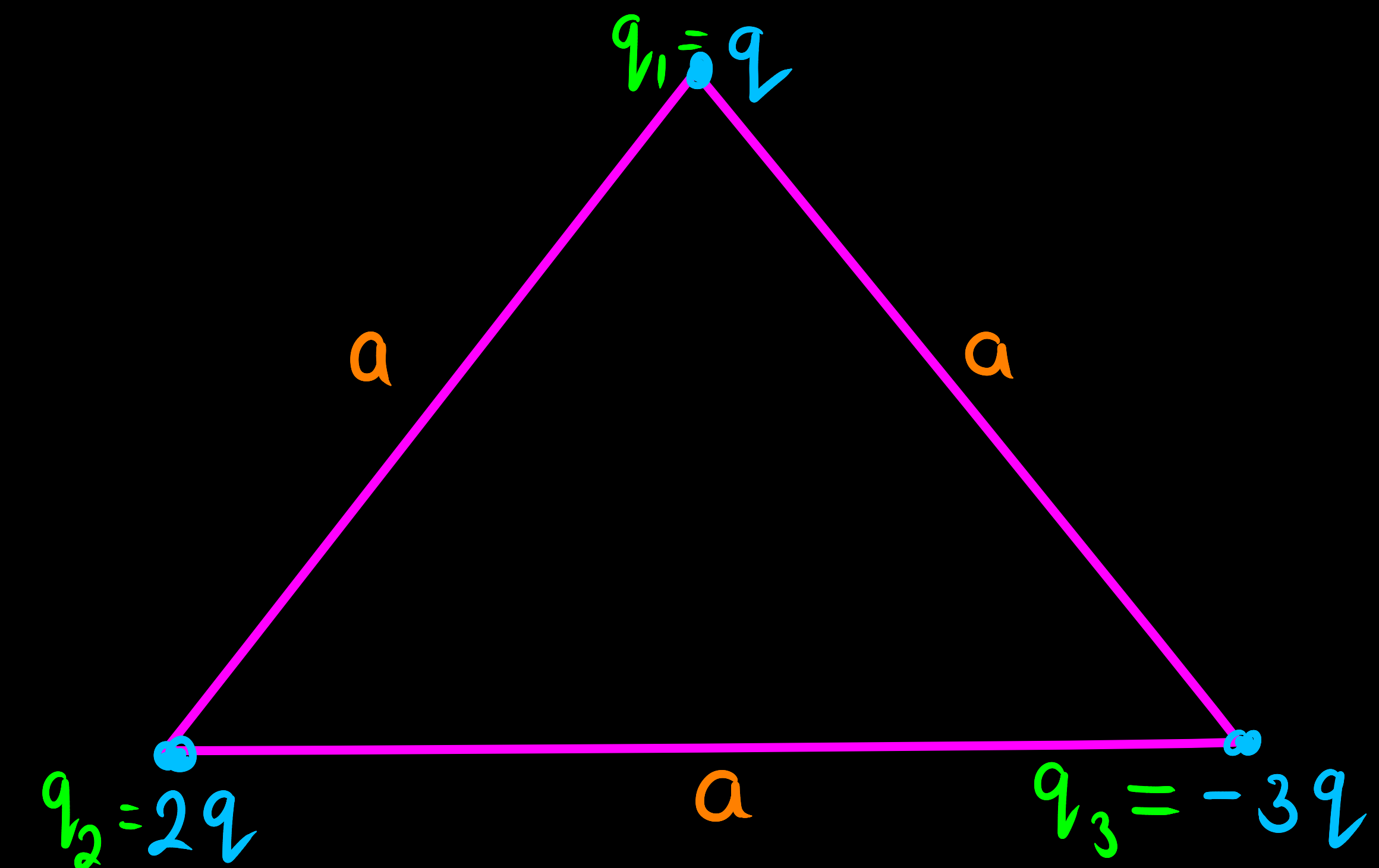
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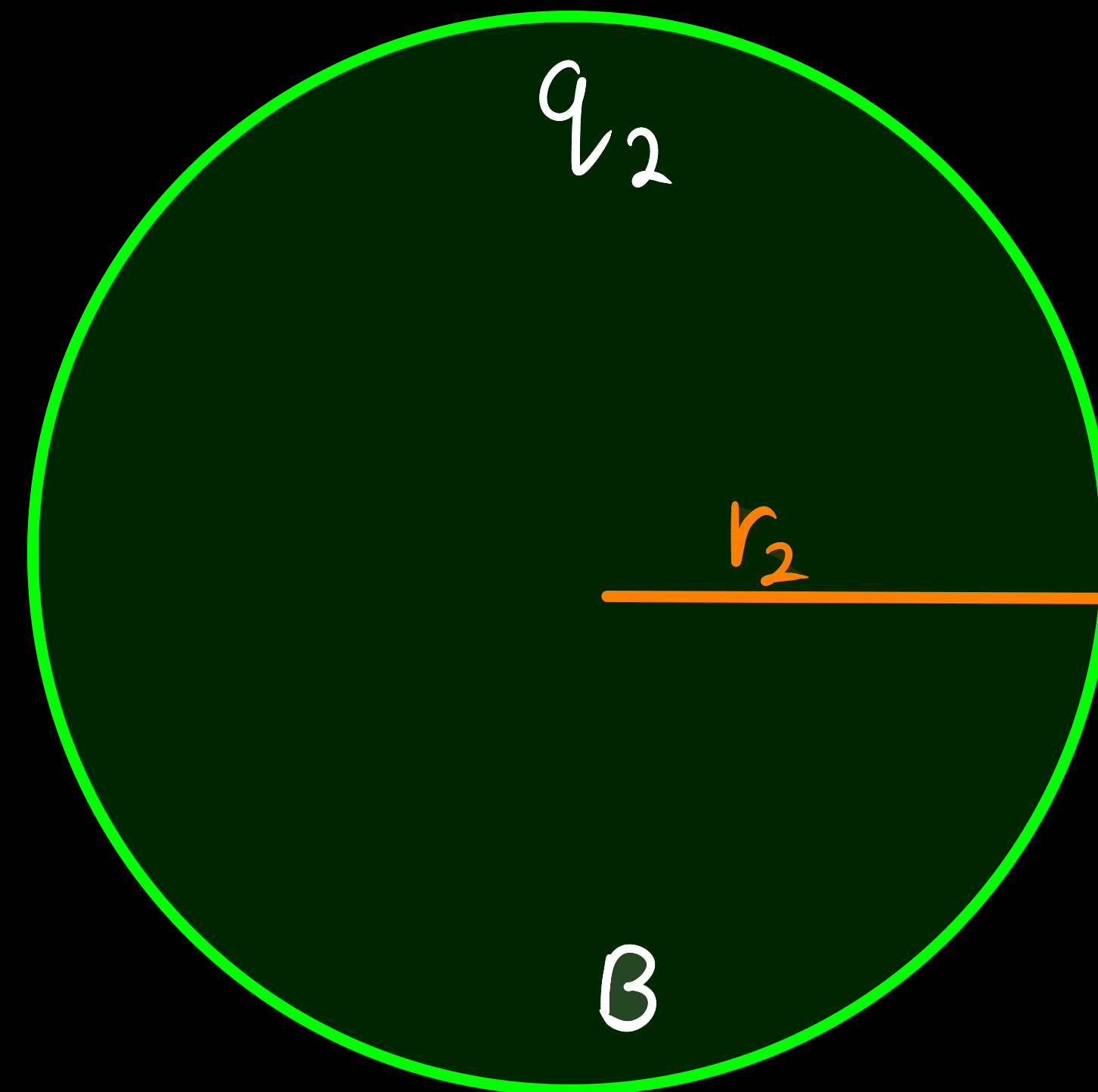
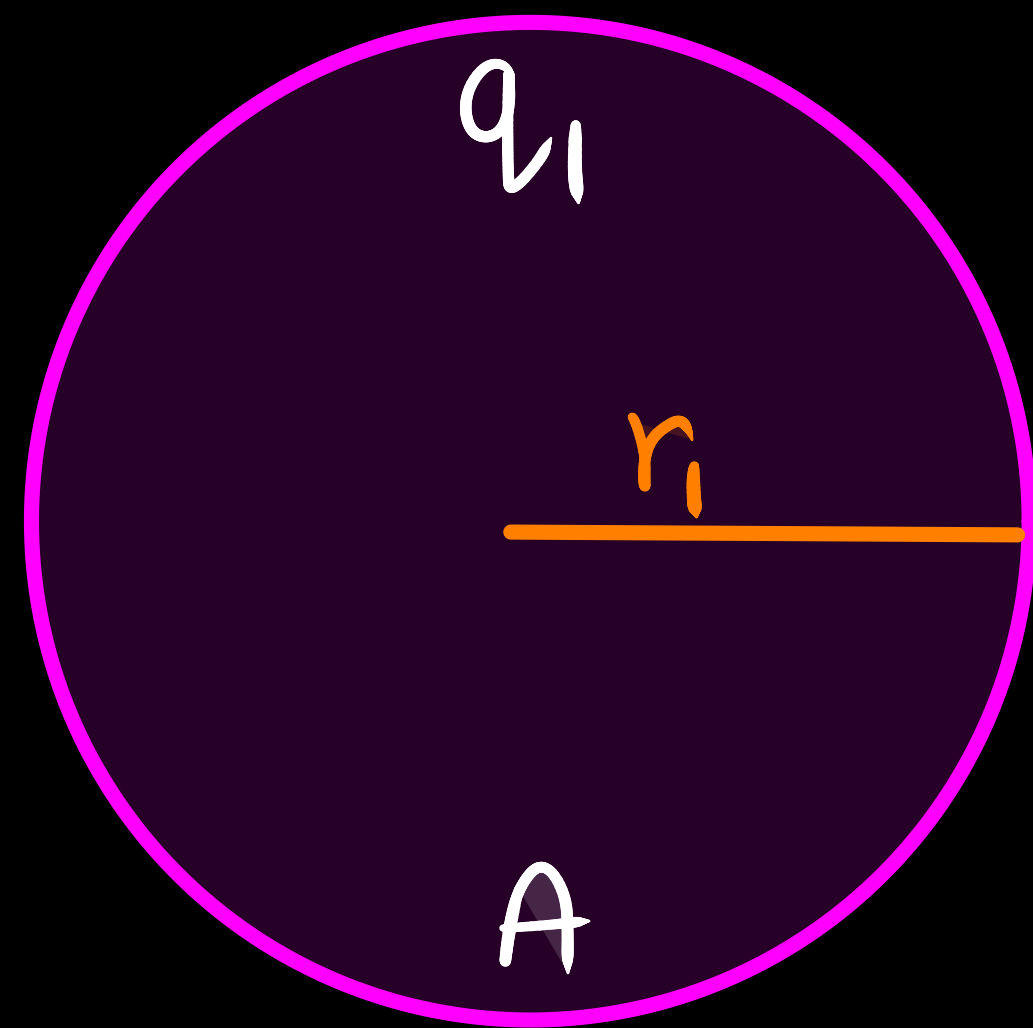
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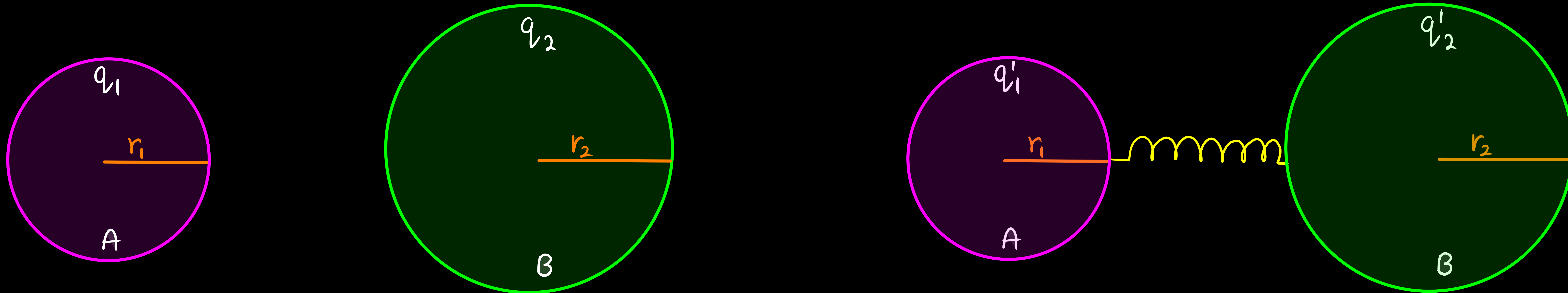
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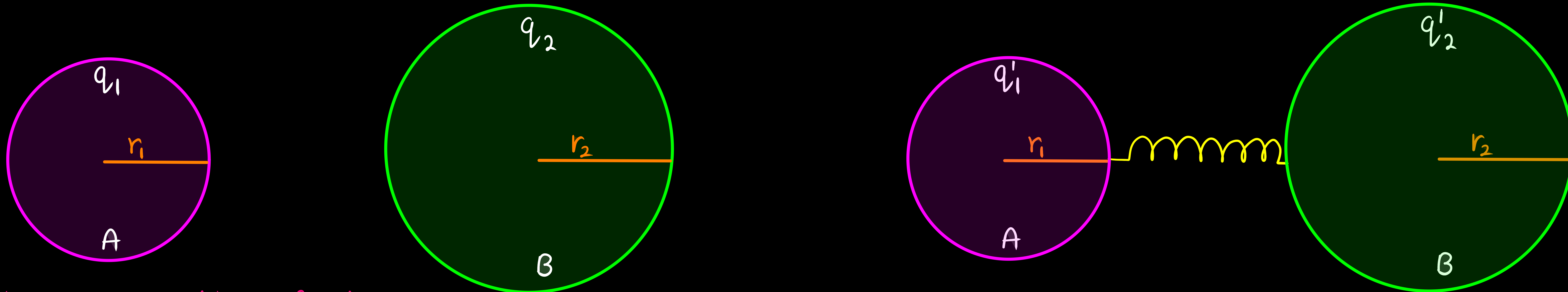
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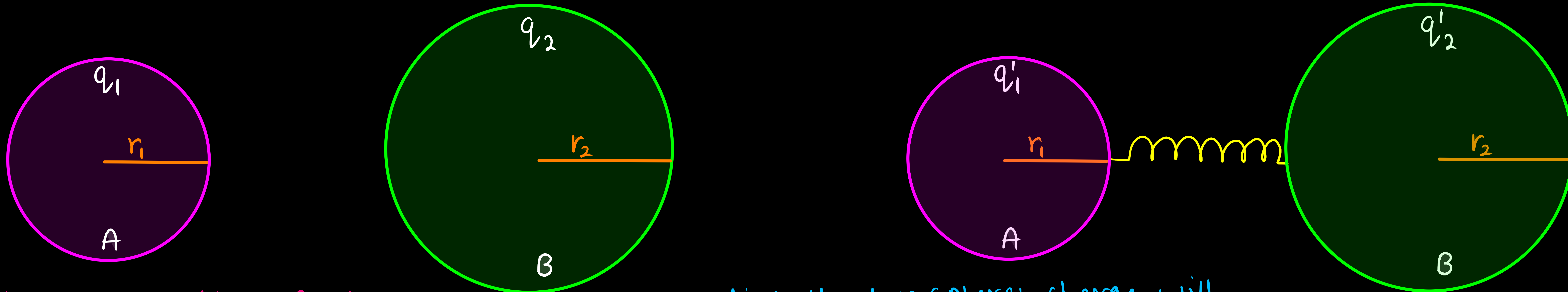
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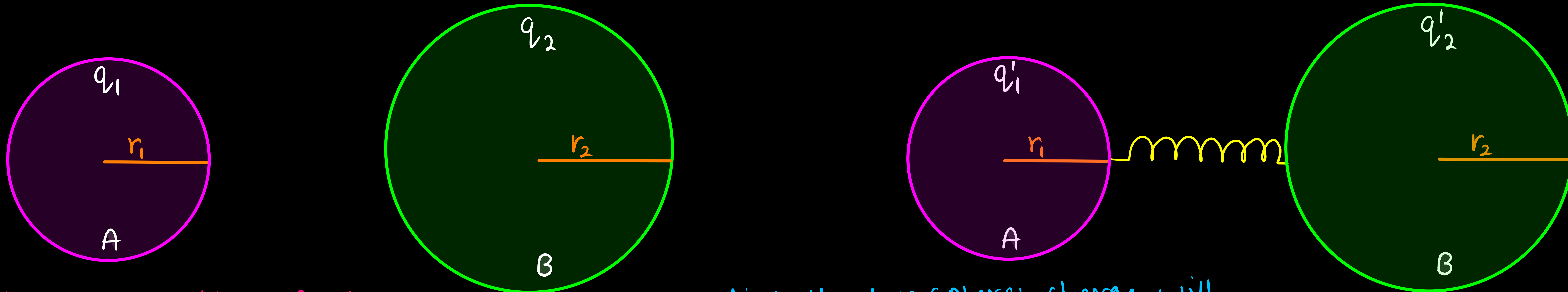
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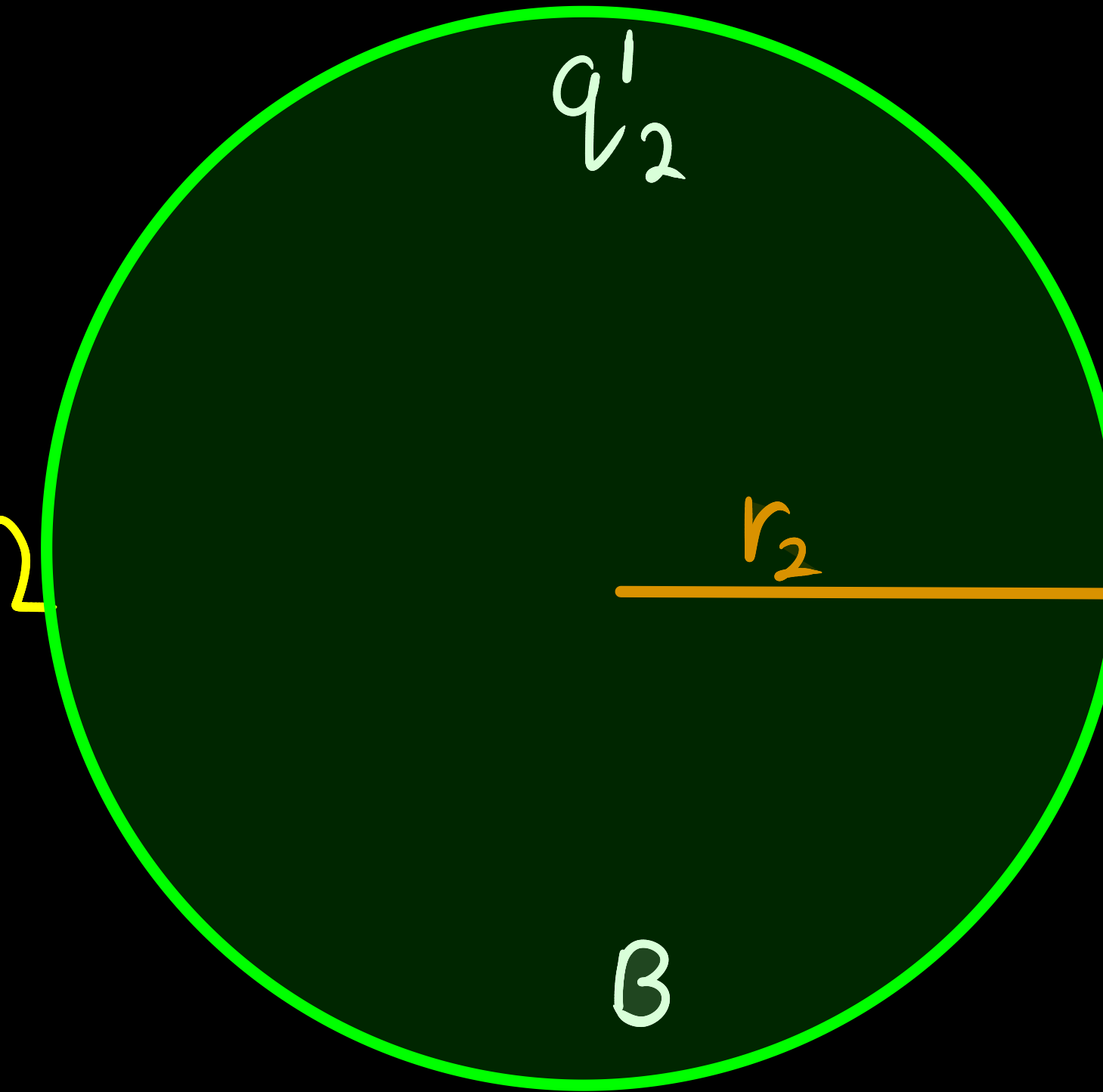
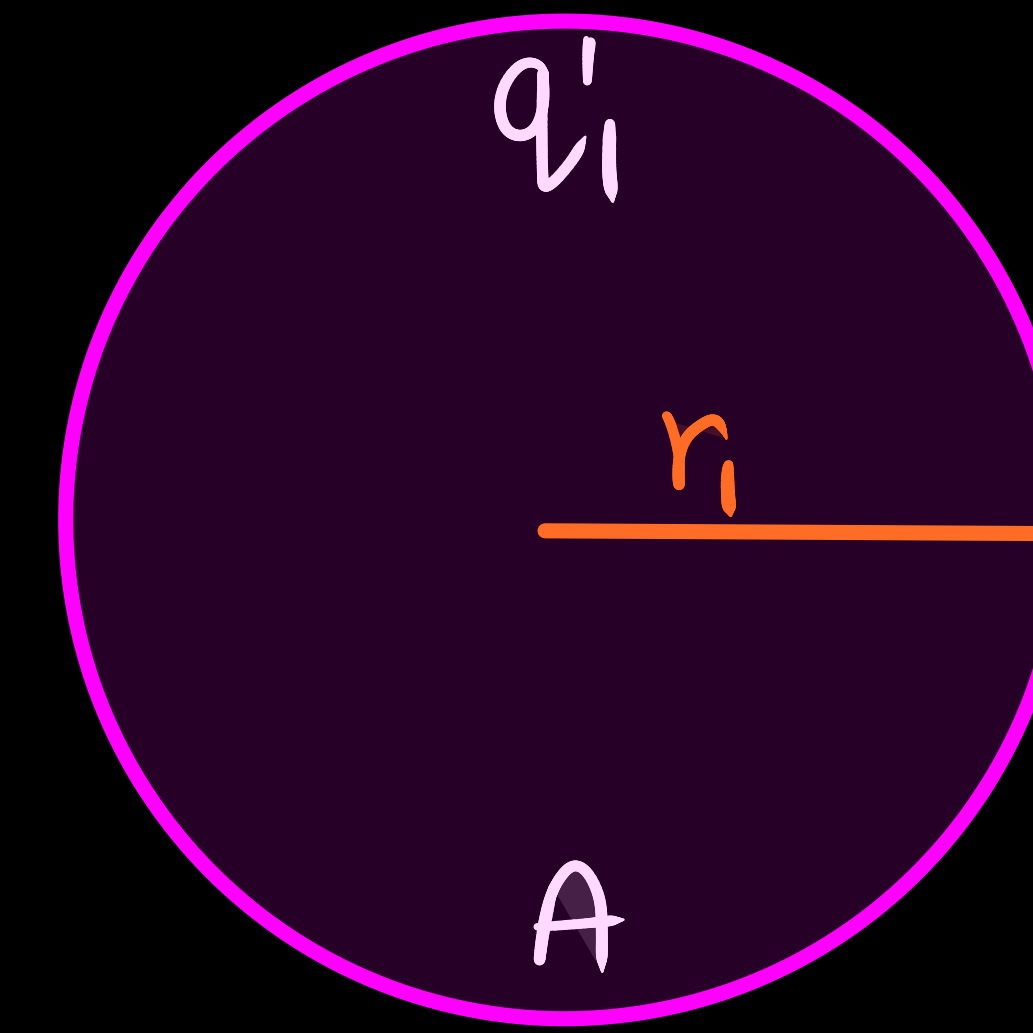
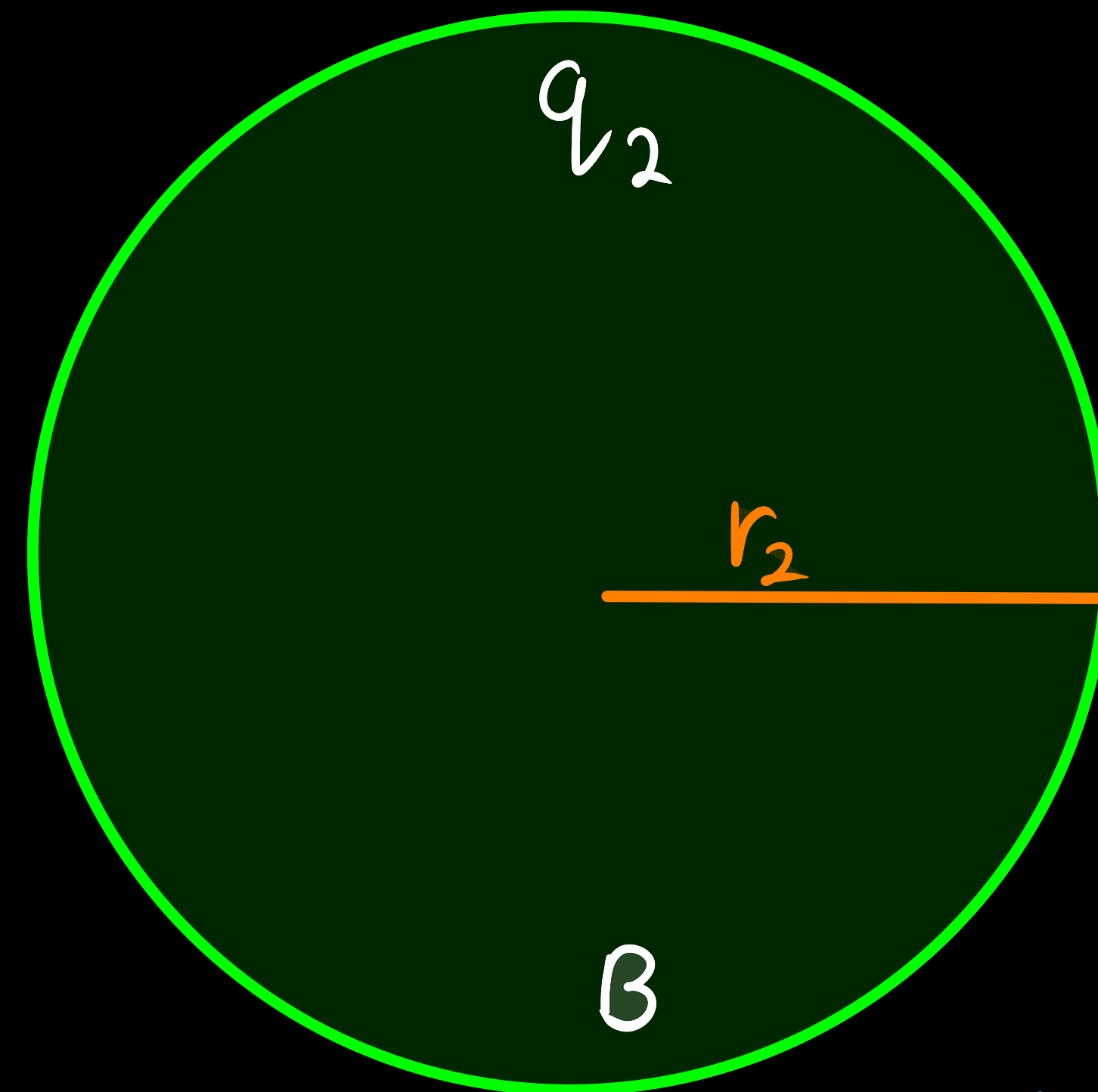
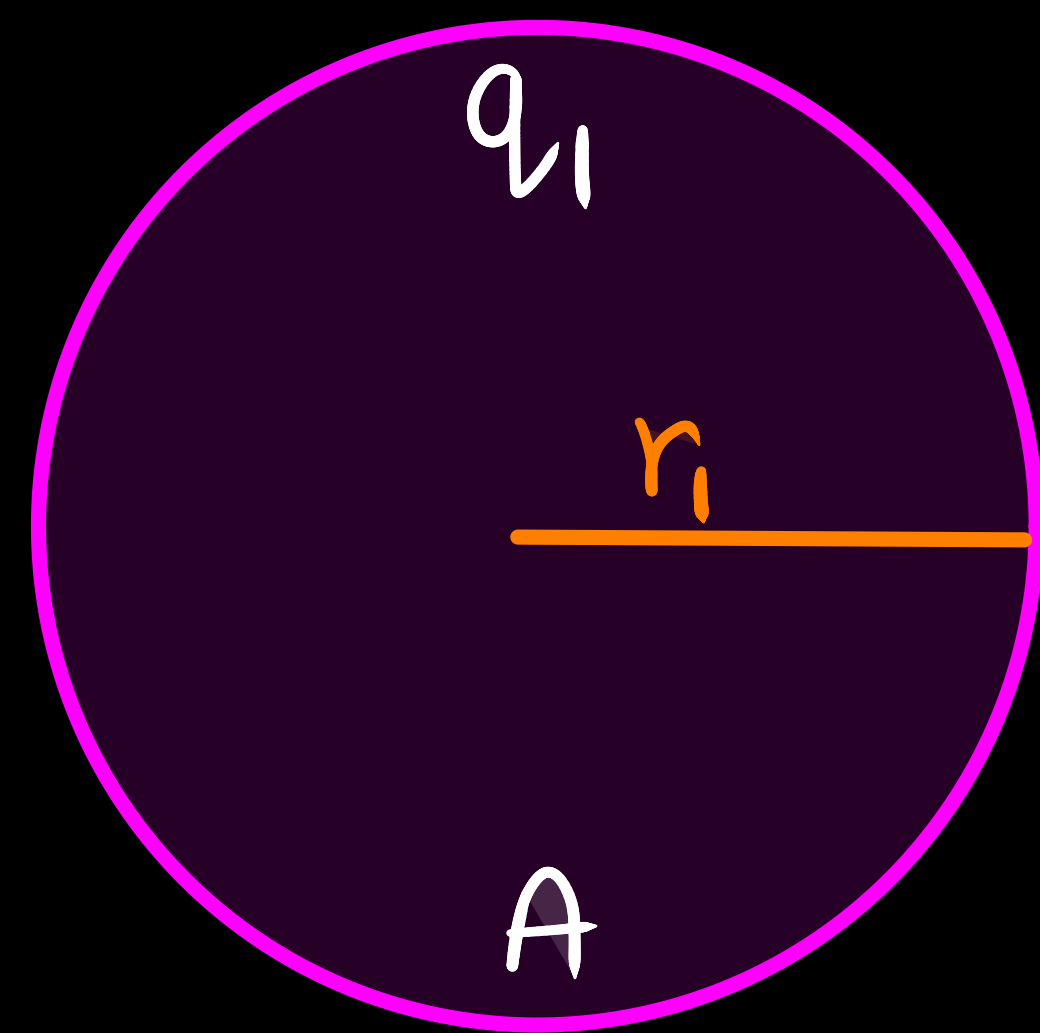
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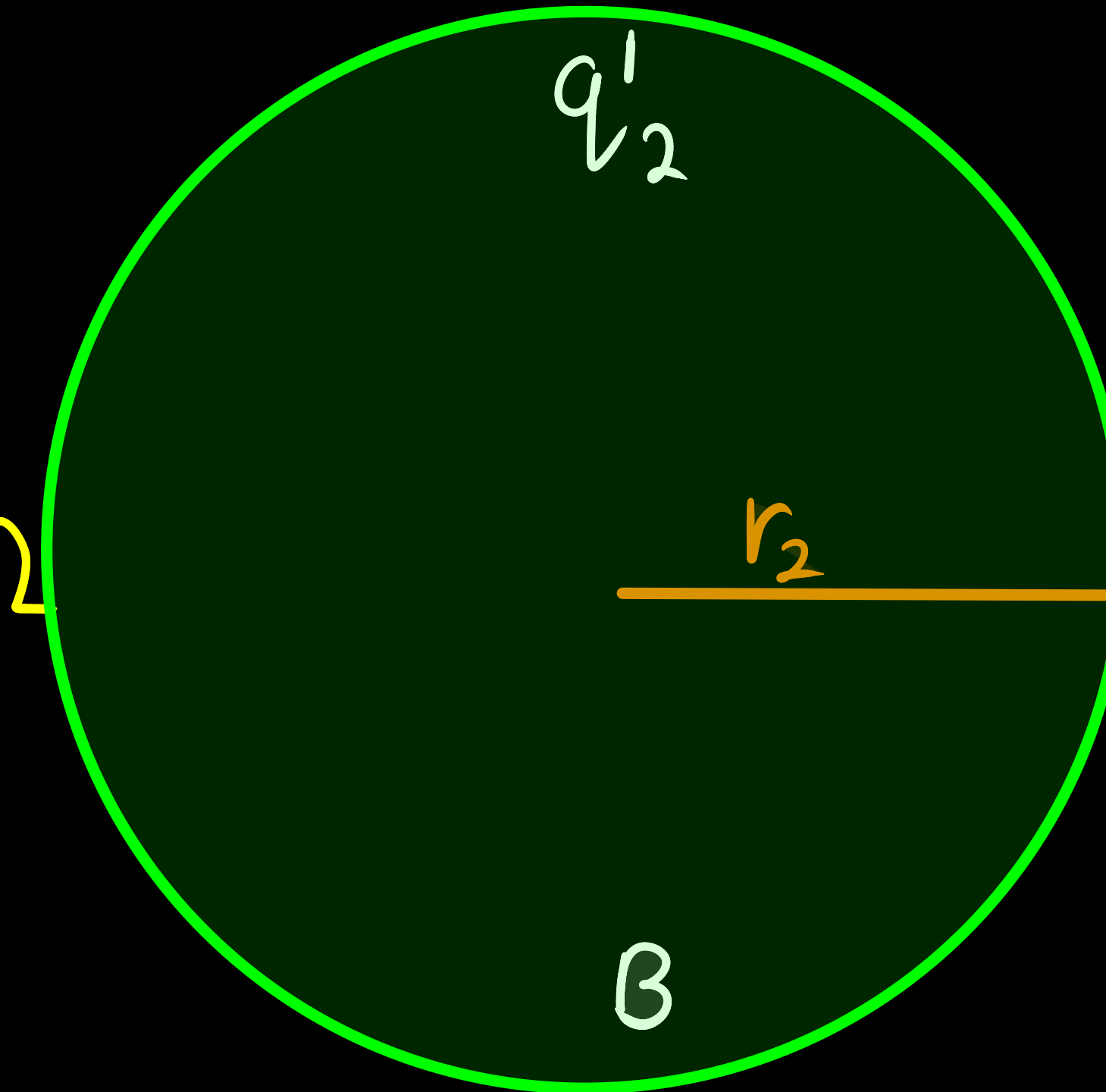
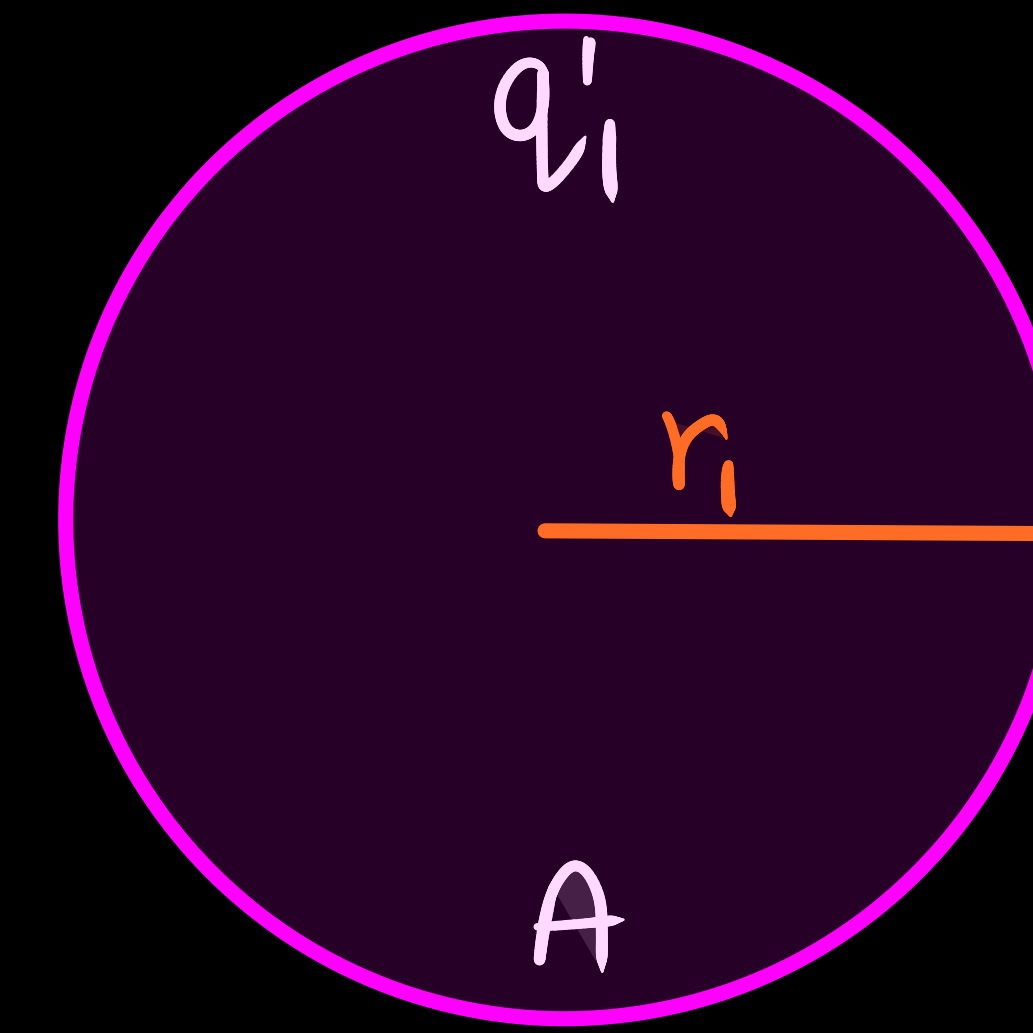
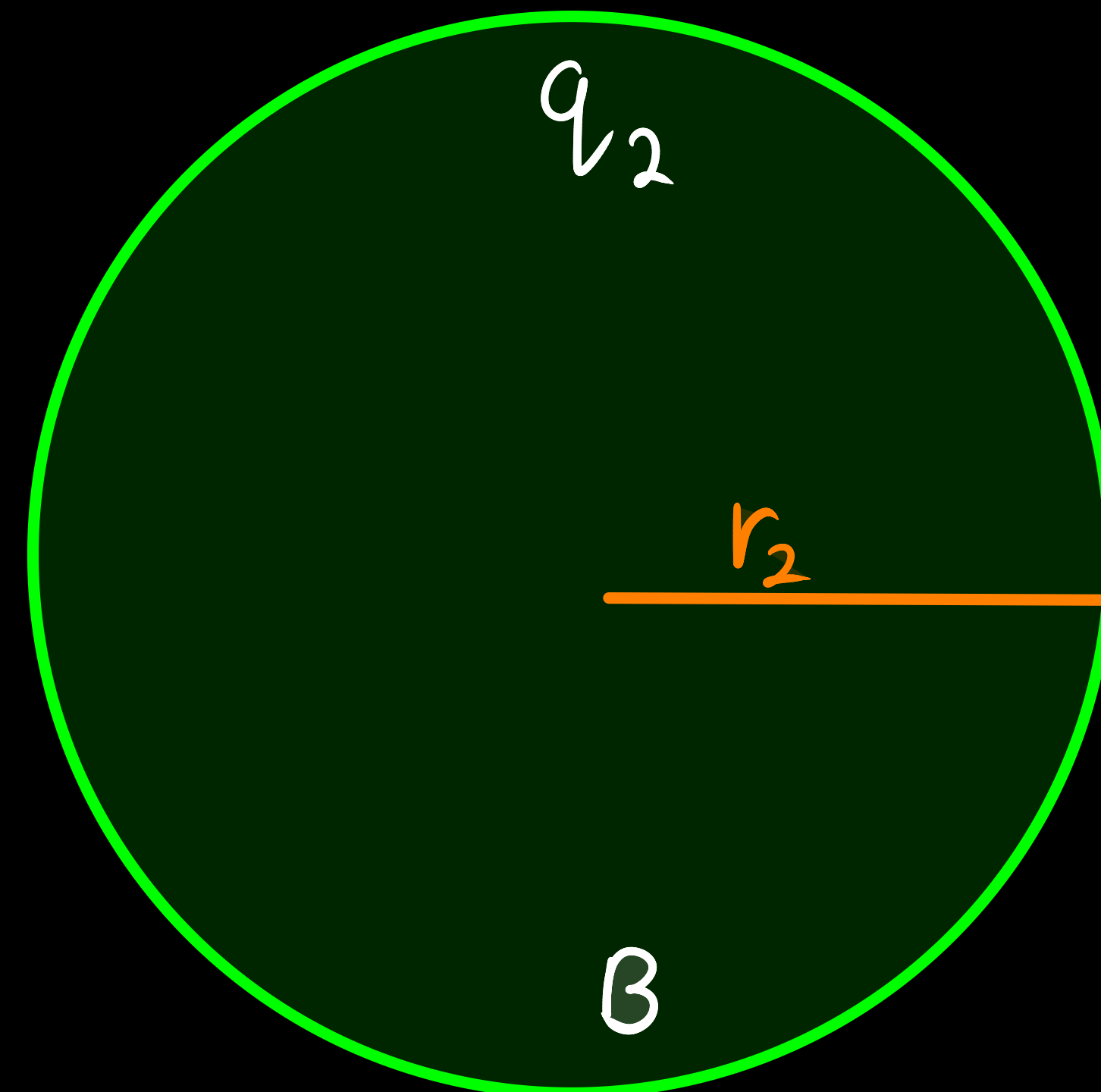
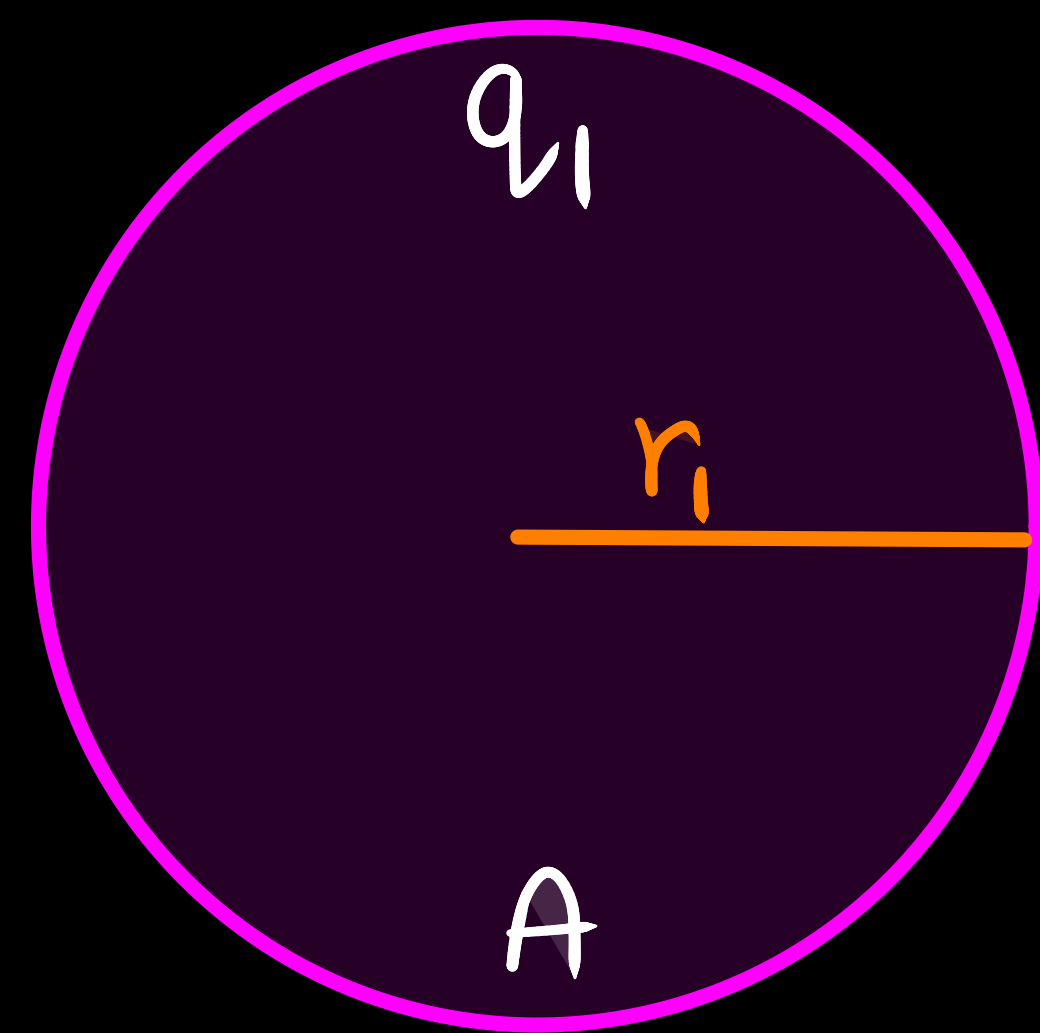
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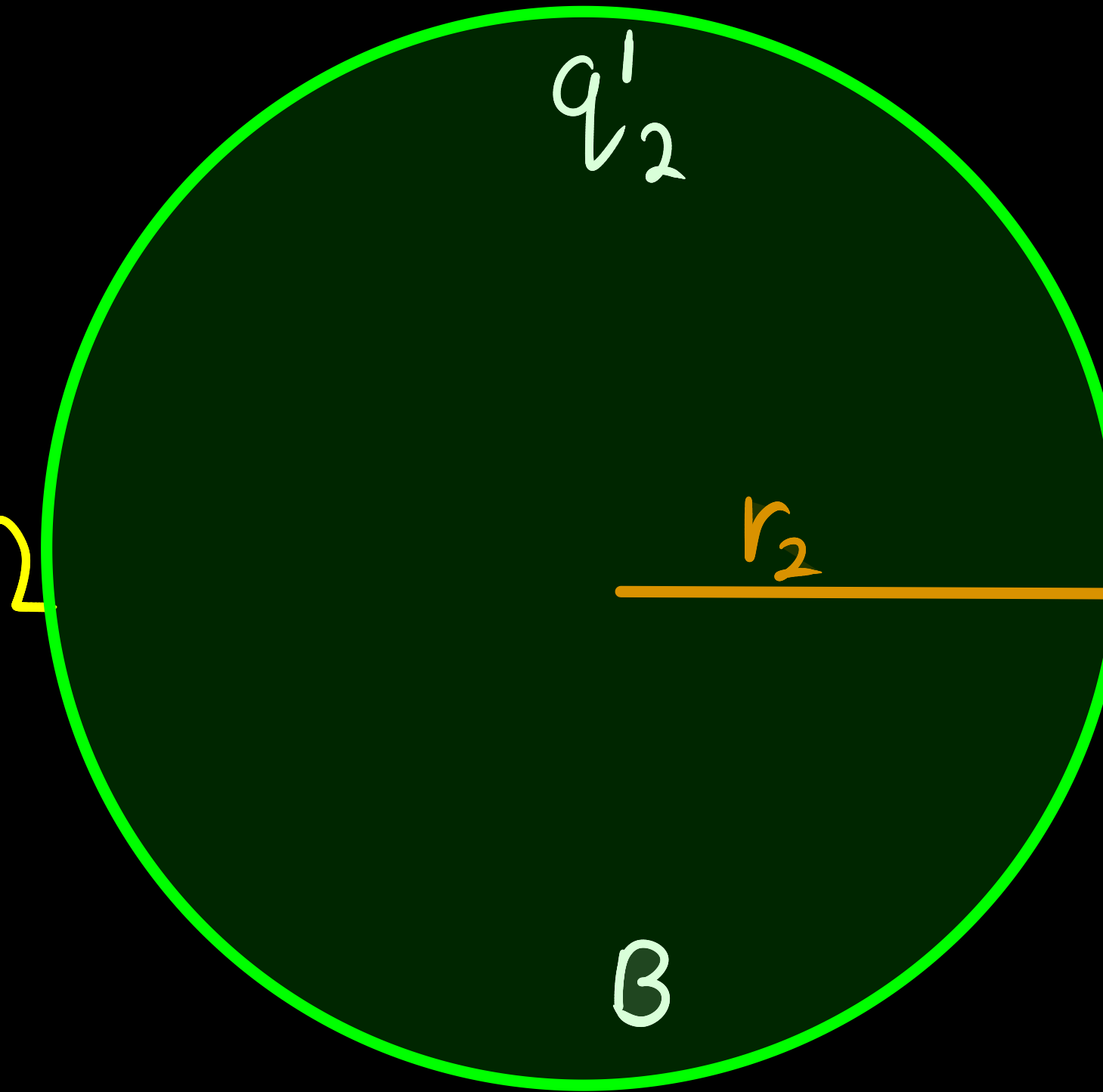
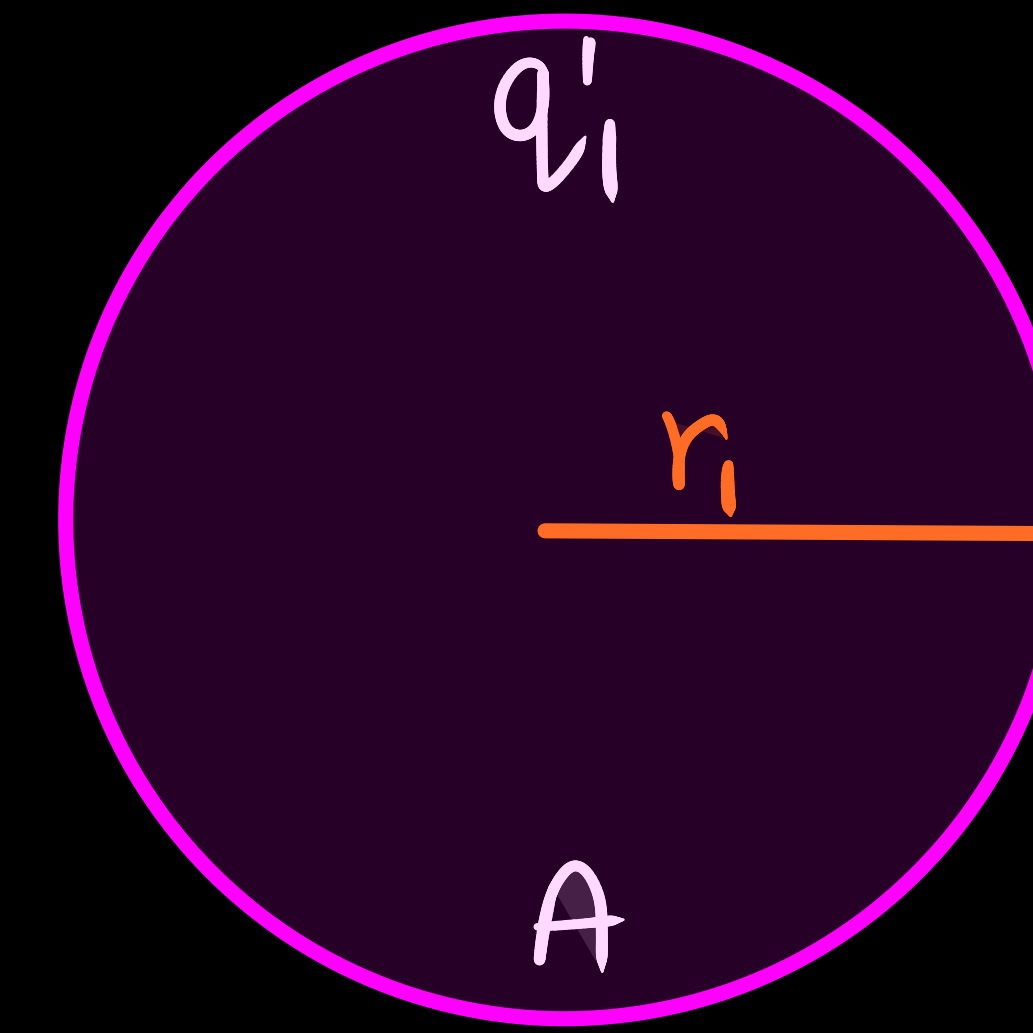
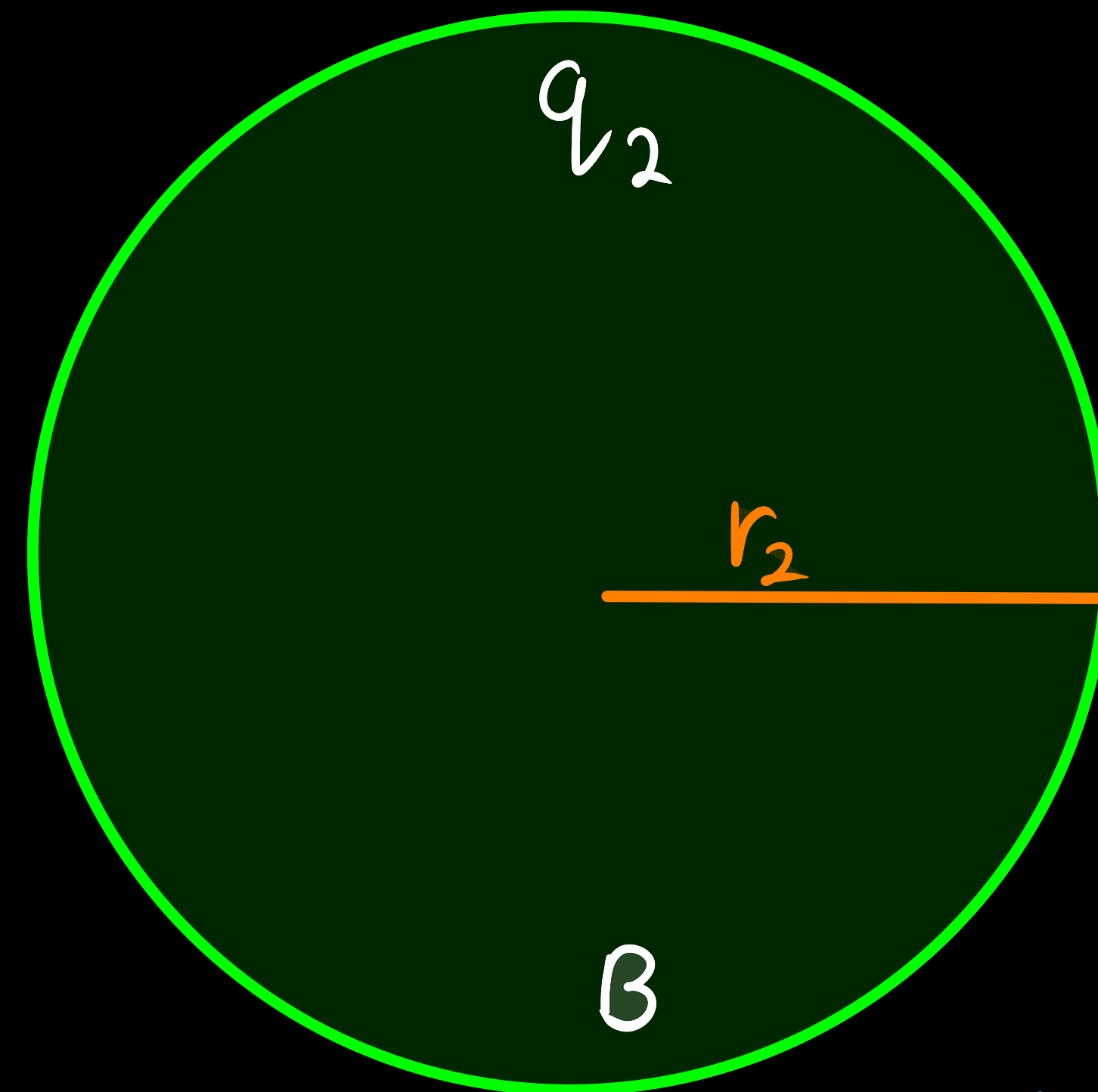
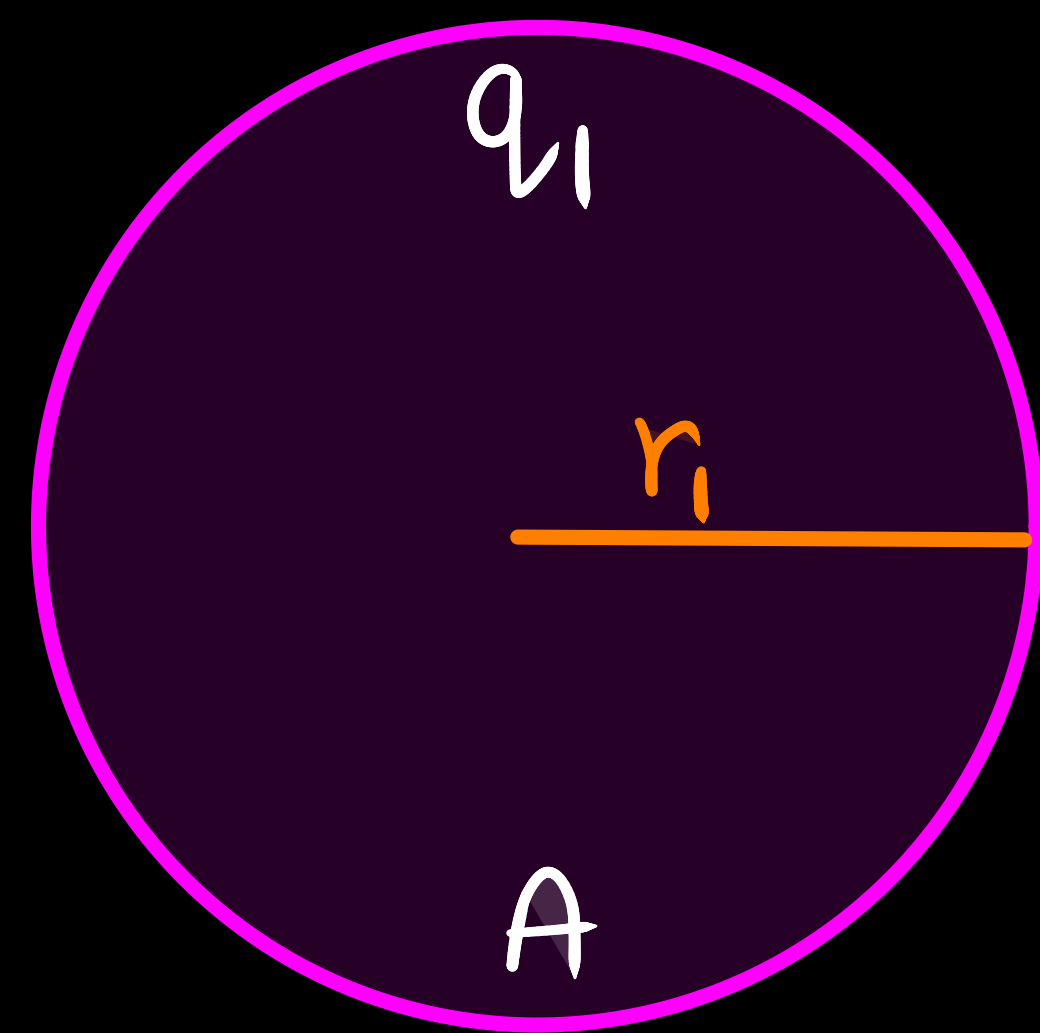
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(B) $-\frac{6q^2}{8\pi\epsilon_0 a}$

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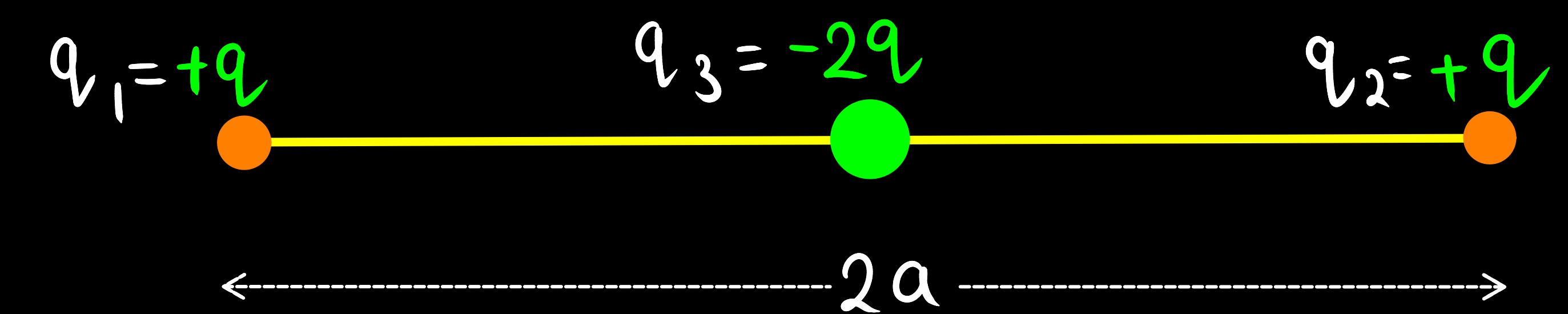
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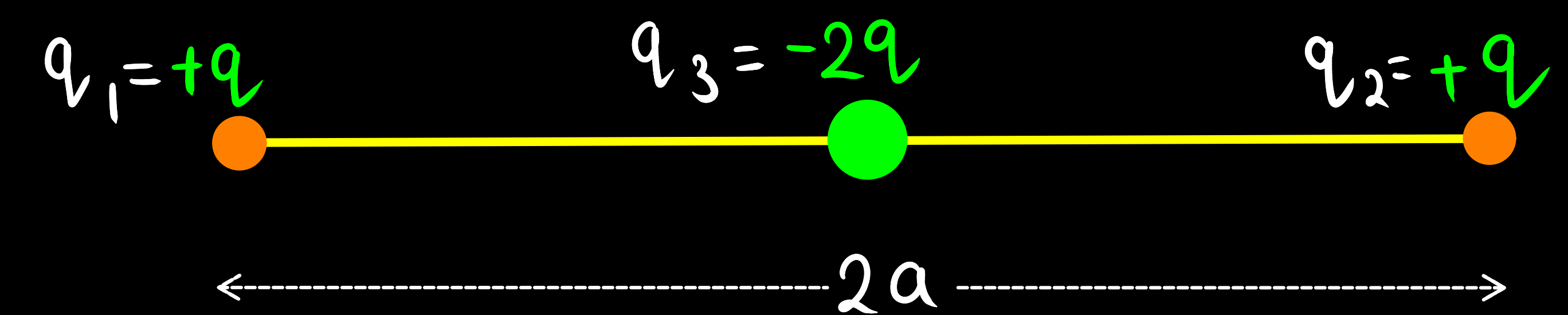
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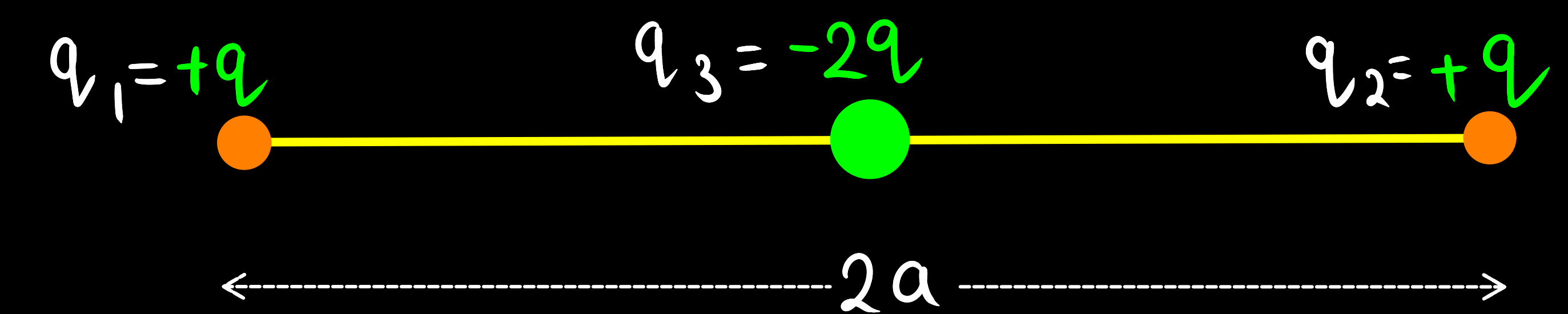
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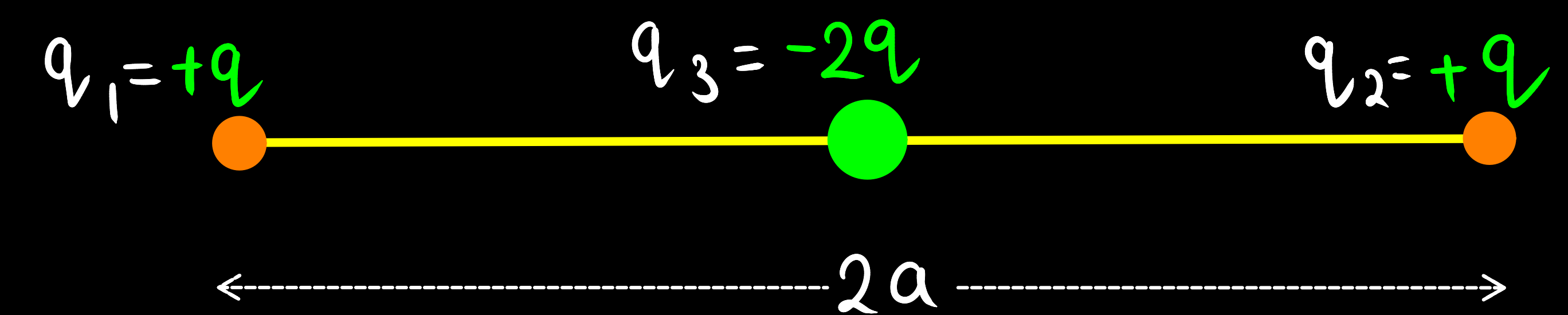
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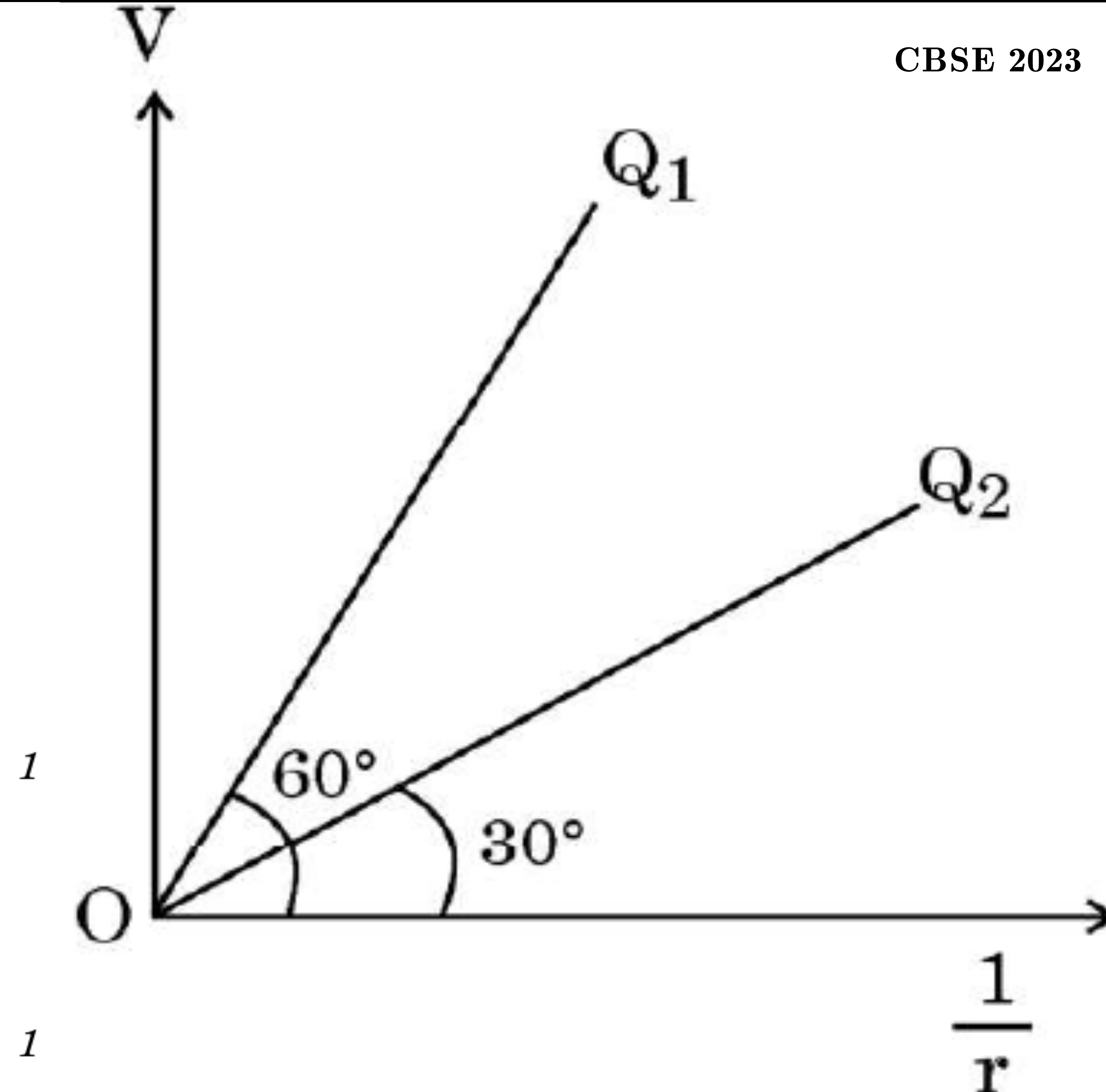
$$U = \frac{1}{4\pi\epsilon_0} \times \frac{(-7q^2)}{2a} = \underline{\underline{\frac{-7q^2}{8\pi\epsilon_0 a}}}$$



34. Electrostatics deals with the study of forces, fields and potentials arising from static charges. Force and electric field, due to a point charge is basically determined by Coulomb's law. For symmetric charge configurations, Gauss's law, which is also based on Coulomb's law, helps us to find the electric field. A charge/a system of charges like a dipole experience a force/torque in an electric field. Work is required to be done to provide a specific orientation to a dipole with respect to an electric field.

Answer the following questions based on the above :

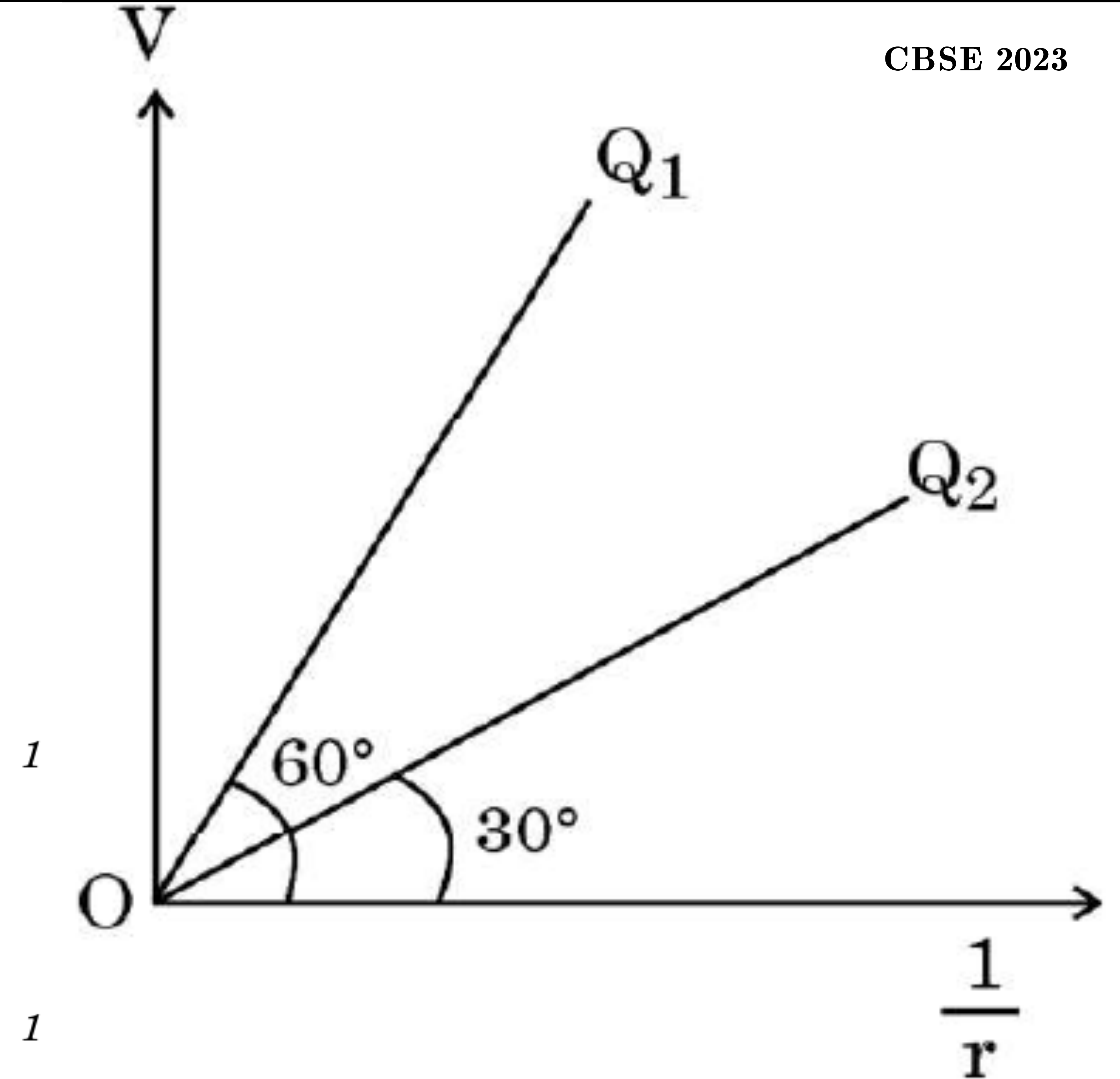
- (a) Consider a uniformly charged thin conducting shell of radius R . Plot a graph showing the variation of $|\vec{E}|$ with distance r from the centre, for points $0 \leq r \leq 3R$. $\rightarrow$ Refer your notebook.
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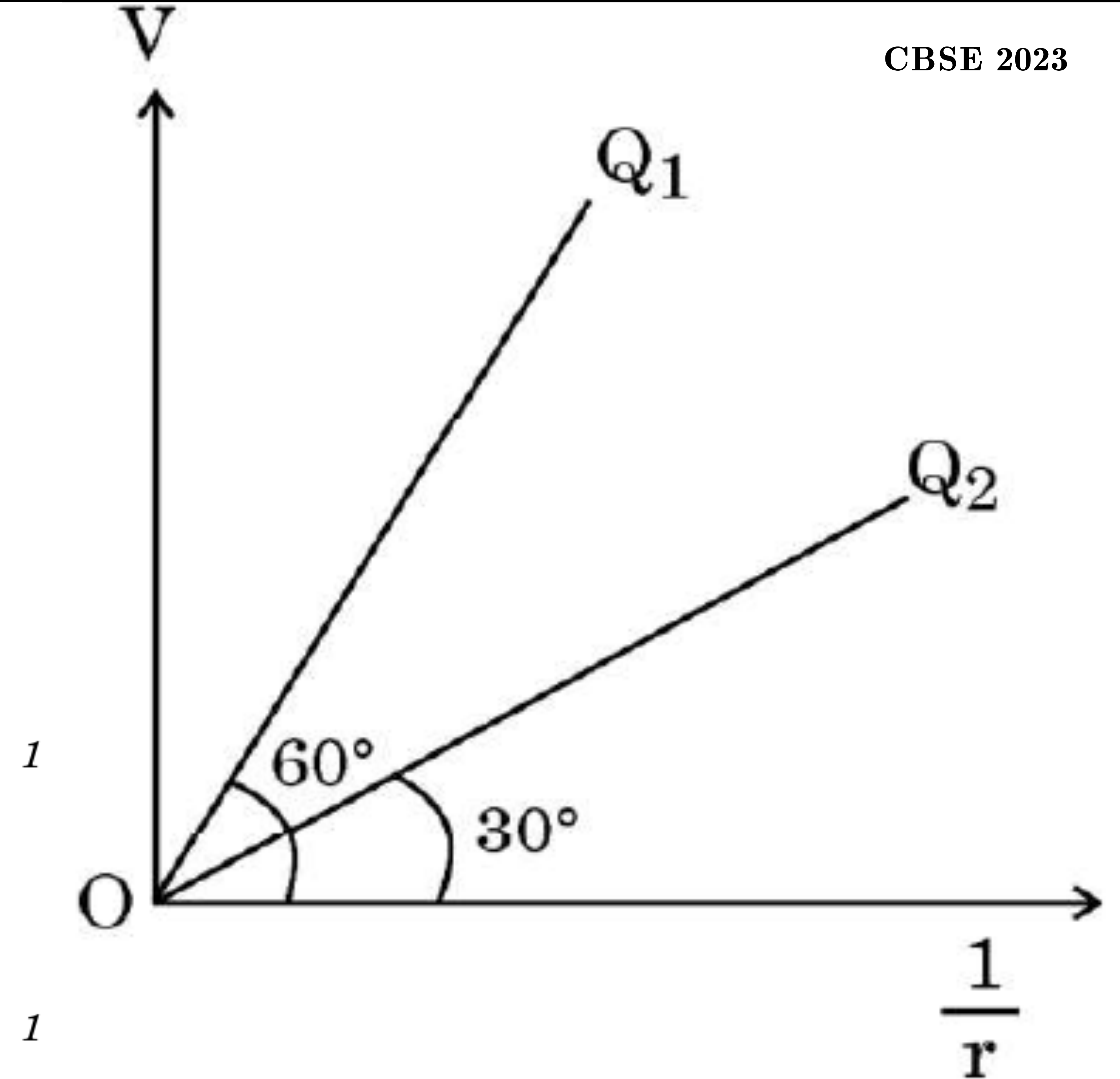
$$V = kQ \times \frac{1}{r}$$

$\downarrow$ $\downarrow$ $\downarrow$
 y $\text{Slope}(m)$ x
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$$V = kQ \times \frac{1}{r}$$

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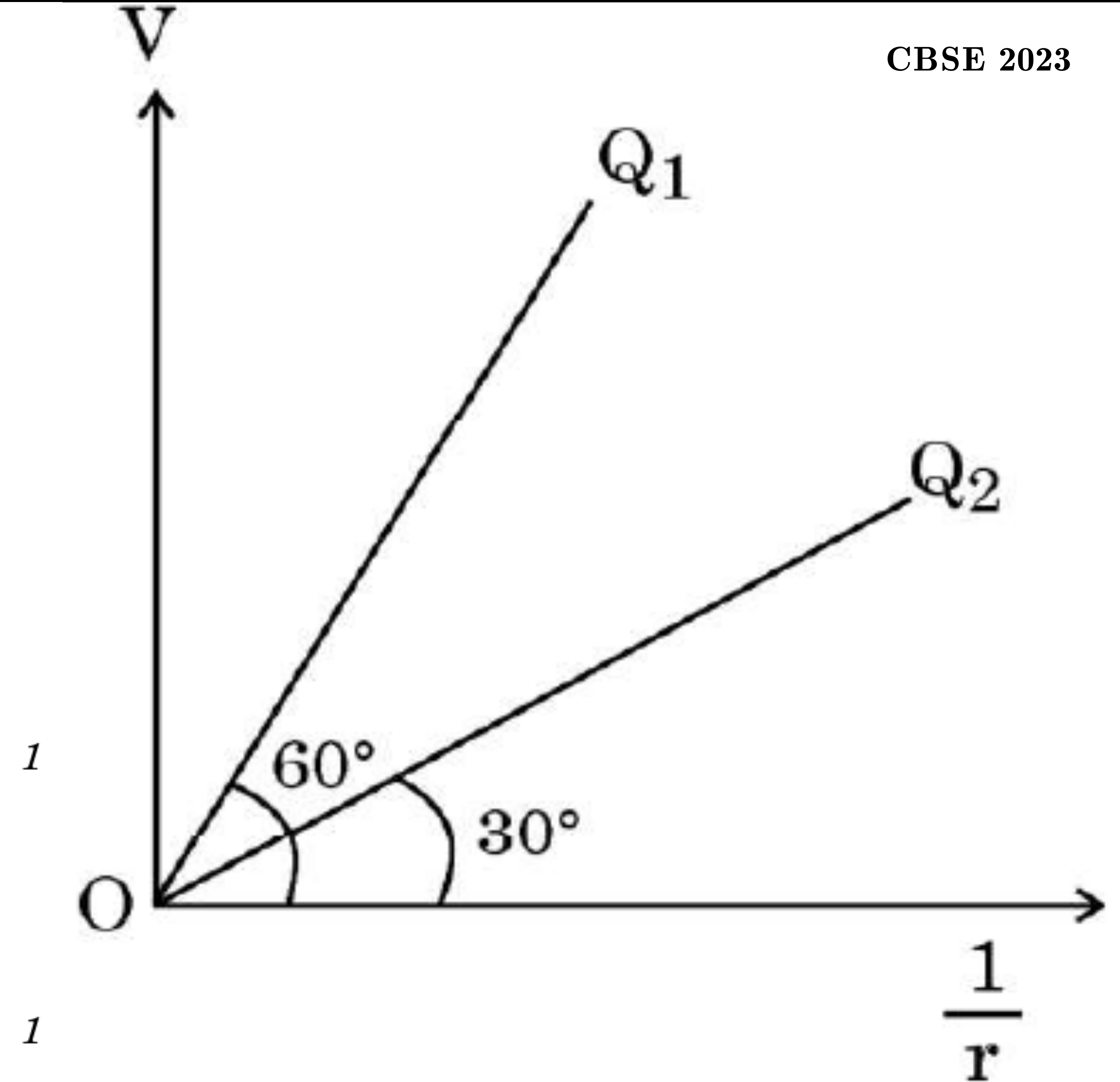
$$V_1 = \frac{kQ_1}{r_1}$$

$$\tan \theta_1 = kQ_1 \quad \text{--- (1)}$$

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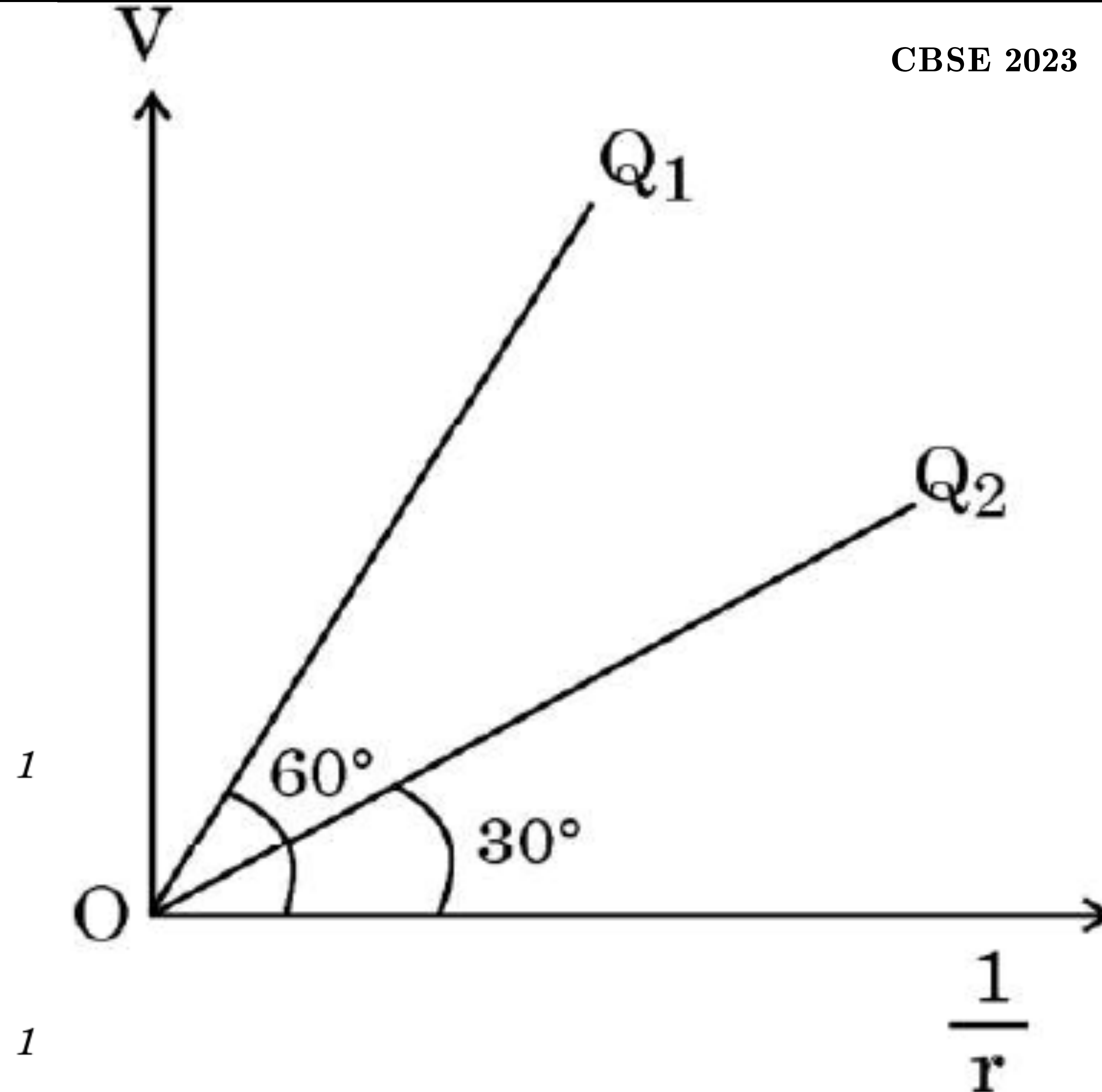
$$V_2 = \frac{kQ_2}{r_2}$$

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b) $V = kQ \times \frac{1}{r}$
 $y = \text{Slope}(m) = \tan(\theta)$
 x

$$V_1 = \frac{kQ_1}{r_1}$$

$$\tan \theta_1 = kQ_1 \quad \text{--- (1)}$$

$$V_2 = \frac{kQ_2}{r_2}$$

$$\tan \theta_2 = kQ_2 \quad \text{--- (2)}$$

Divide eqⁿ (1) by (2)

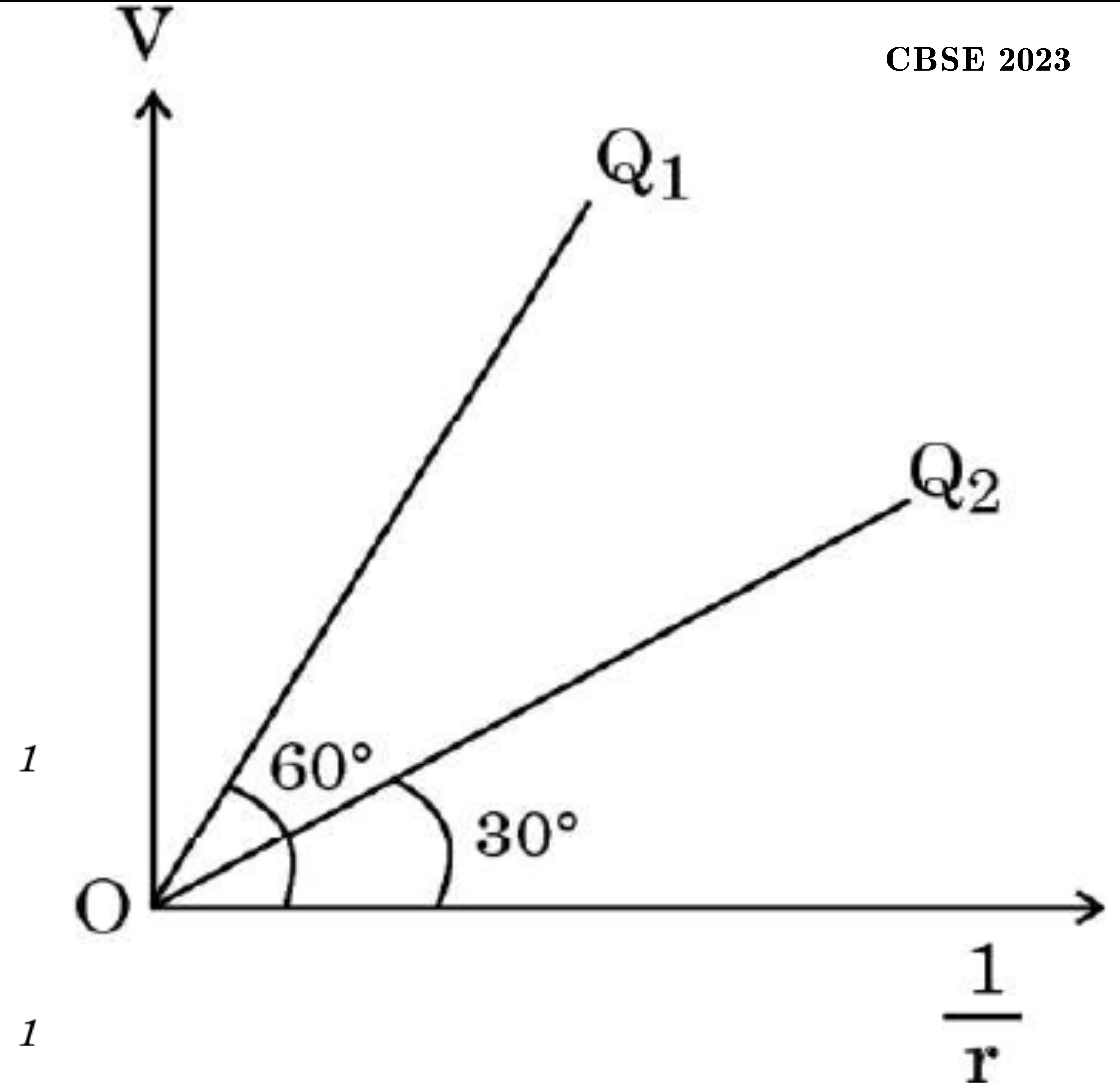
$$\frac{kQ_1}{kQ_2} = \frac{\tan \theta_1}{\tan \theta_2}$$

$$\frac{Q_1}{Q_2} = \frac{\tan 60^\circ}{\tan 30^\circ}$$

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b) $V = kQ \times \frac{1}{r}$
 $y = \text{Slope}(m) = \tan(\theta)$
 x

$$V_1 = \frac{kQ_1}{r_1}$$

$$\tan \theta_1 = kQ_1 \quad \text{--- (1)}$$

$$V_2 = \frac{kQ_2}{r_2}$$

$$\tan \theta_2 = kQ_2 \quad \text{--- (2)}$$

Divide eqⁿ (1) by (2)

$$\frac{kQ_1}{kQ_2} = \frac{\tan \theta_1}{\tan \theta_2}$$

$$\frac{Q_1}{Q_2} = \frac{\tan 60^\circ}{\tan 30^\circ}$$

$$\frac{Q_1}{Q_2} = \frac{\sqrt{3}}{\frac{1}{\sqrt{3}}}$$

$$\boxed{\frac{Q_1}{Q_2} = 3}$$

- (c) An electric dipole of dipole moment of 6×10^{-7} C-m is kept in a uniform electric field of 10^4 N/C such that the dipole moment and the electric field are parallel. Calculate the potential energy of the dipole.

2

OR

- (c) An electric dipole of dipole moment $\vec{p}$ is initially kept in a uniform electric field $\vec{E}$ such that $\vec{p}$ is perpendicular to $\vec{E}$. Find the amount of work done in rotating the dipole to a position at which $\vec{p}$ becomes antiparallel to $\vec{E}$.

CBSE 2023

2

c) i) $p = 6 \times 10^{-7} \text{ C-m}$
 $E = 10^4 \text{ N/C}$
 $\theta = 0^\circ$

- (c) An electric dipole of dipole moment of $6 \times 10^{-7} \text{ C-m}$ is kept in a uniform electric field of 10^4 N/C such that the **dipole moment and the electric field are parallel**. Calculate the potential energy of the dipole.

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2

c) i) $p = 6 \times 10^{-7} \text{ C-m}$
 $E = 10^4 \text{ N/C}$
 $\theta = 0^\circ$

$$U = -pE \cos \theta$$

$$U = -6 \times 10^{-7} \times 10^4 \times \cos(0)$$

$$U = -6 \times 10^{-3} \text{ J}$$

- (c) An electric dipole of dipole moment of $6 \times 10^{-7} \text{ C-m}$ is kept in a uniform electric field of 10^4 N/C such that the **dipole moment and the electric field are parallel**. Calculate the potential energy of the dipole.

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c) ii) $\theta_i = 90^\circ$
 $\theta_f = 180^\circ$

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2

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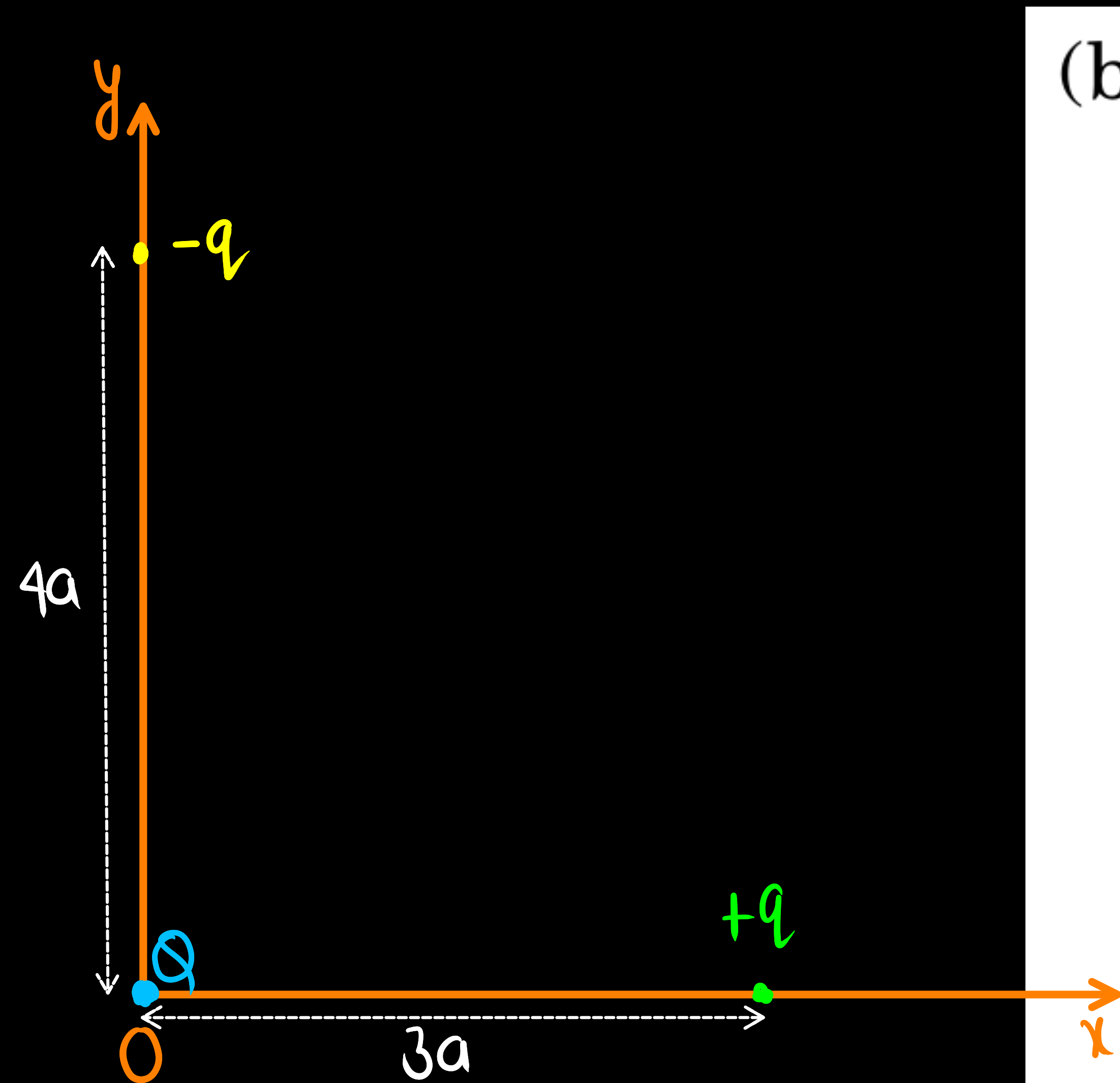
2

c) ii) $\theta_i = 90^\circ$
 $\theta_f = 180^\circ$
 $\Delta U = -pE (\cos \theta_f - \cos \theta_i)$
 $\Delta U = -pE (\cos 180^\circ - \cos 90^\circ)$
 $\Delta U = -pE (-1 - 0)$

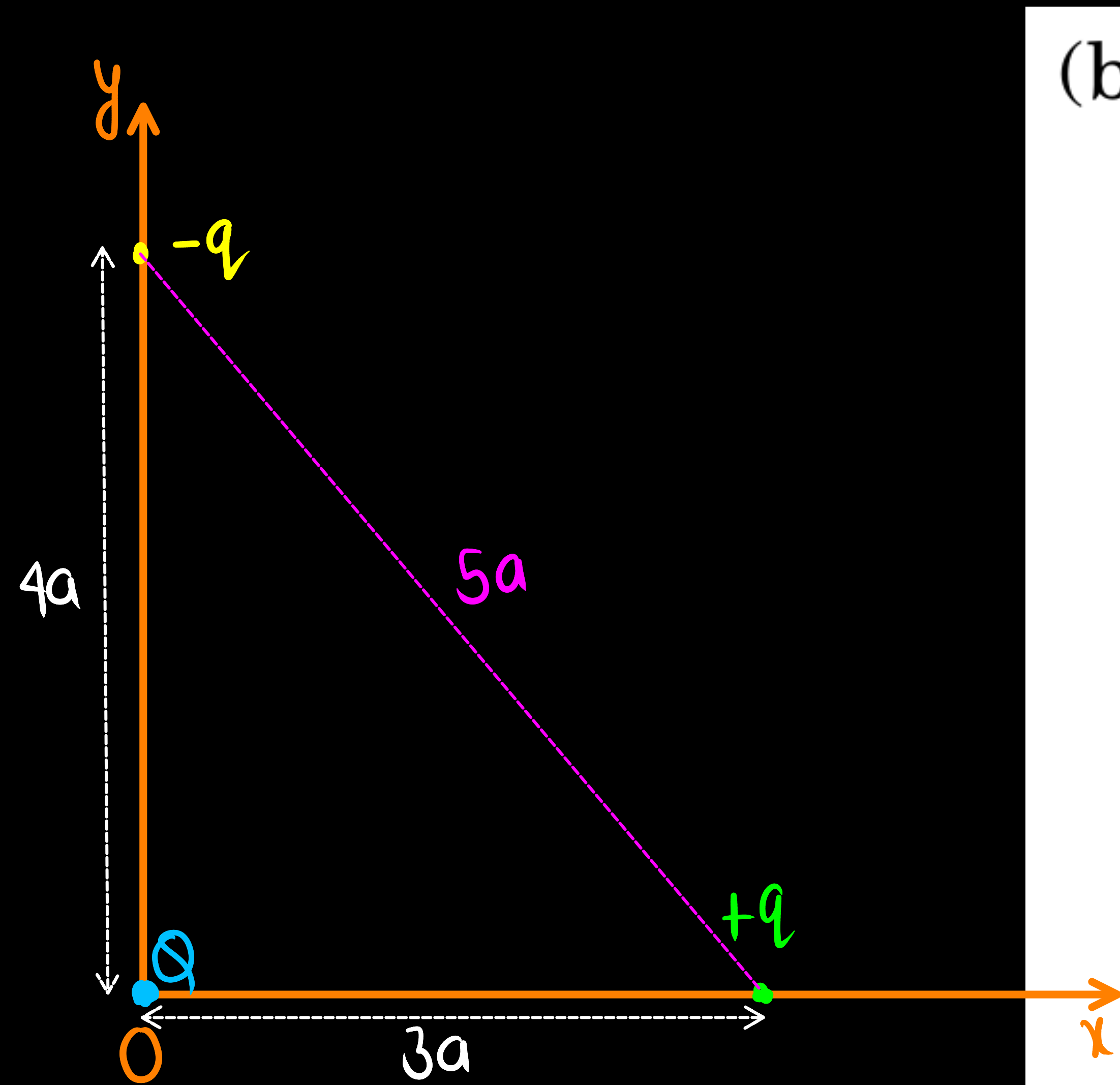
$$\Delta U = pE = W$$

- (b) (i) Define electric potential at a point and write its SI unit.
- (ii) Two capacitors are connected in series. Derive an expression of the equivalent capacitance of the combination.
- (iii) Two point charges $+q$ and $-q$ are located at points $(3a, 0)$ and $(0, 4a)$ respectively in x - y plane. A third charge Q is kept at the origin. Find the value of Q , in terms of q and a , so that the electrostatic potential energy of the system is zero.

- (b) (i) Define electric potential at a point and write its SI unit. $\hookrightarrow \frac{W_{\text{ext}}}{q}$ $\hookrightarrow \frac{J}{C}$ or volt
- (ii) Two capacitors are connected in series. Derive an expression of the equivalent capacitance of the combination. $\frac{1}{C_s} = \frac{1}{C_1} + \frac{1}{C_2}$
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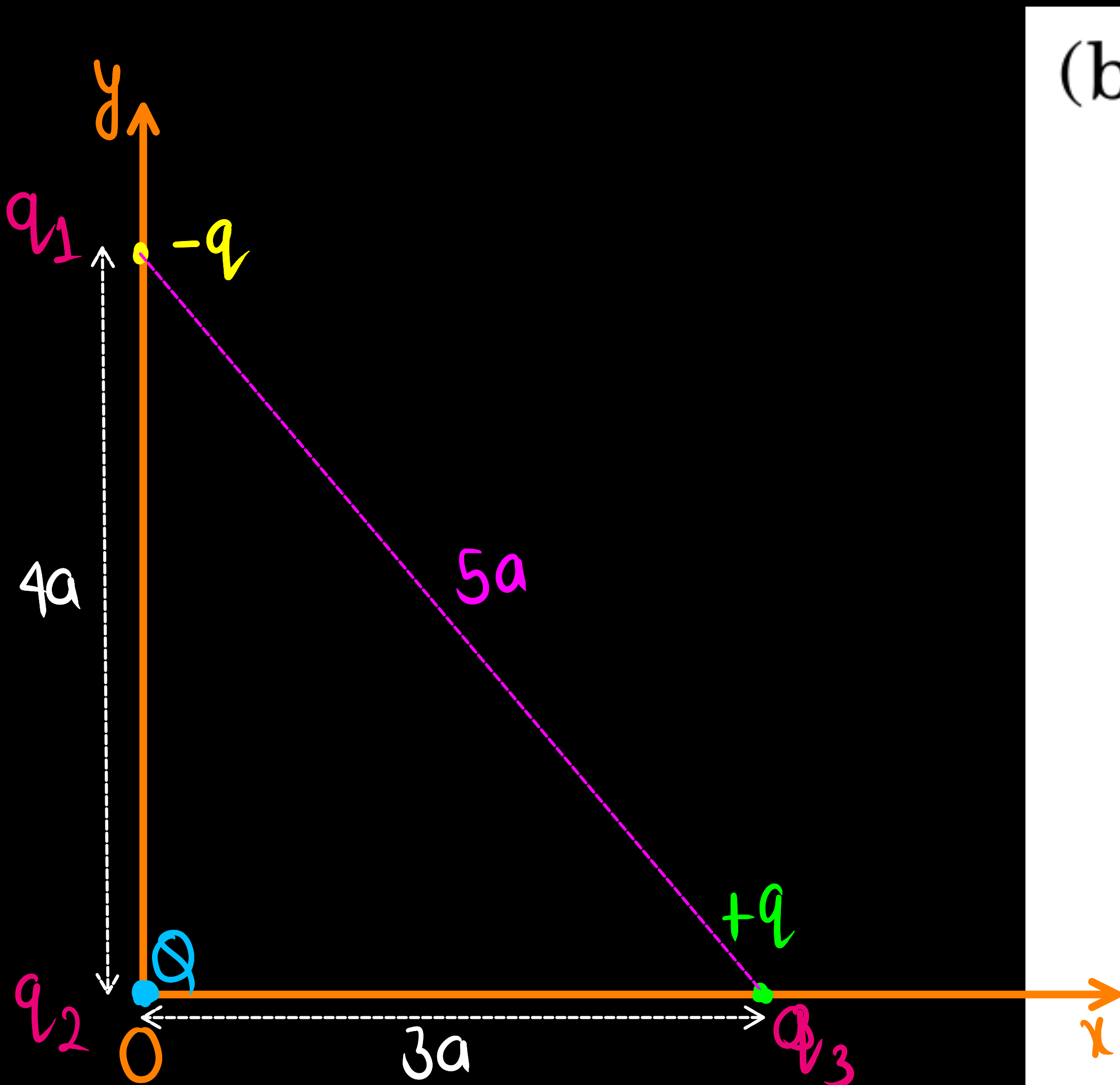


- (b) (i) Define electric potential at a point and write its SI unit.
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Given - $U = 0$

- (b) (i) Define electric potential at a point and write its SI unit.
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Given - $U = 0$

$$\frac{1}{4\pi\epsilon_0} \left(\frac{q_1 q_2}{r_{12}} + \frac{q_1 q_3}{r_{13}} + \frac{q_2 q_3}{r_{23}} \right) = 0$$

$$\left(-\frac{qQ}{4a} - \frac{q^2}{5a} + \frac{qQ}{3a} \right) = 0$$

$$\frac{qQ}{3a} - \frac{qQ}{4a} = \frac{q^2}{5a}$$

(b) (i) Define electric potential at a point and write its SI unit.

$$\hookrightarrow \frac{\text{Work}}{q}$$

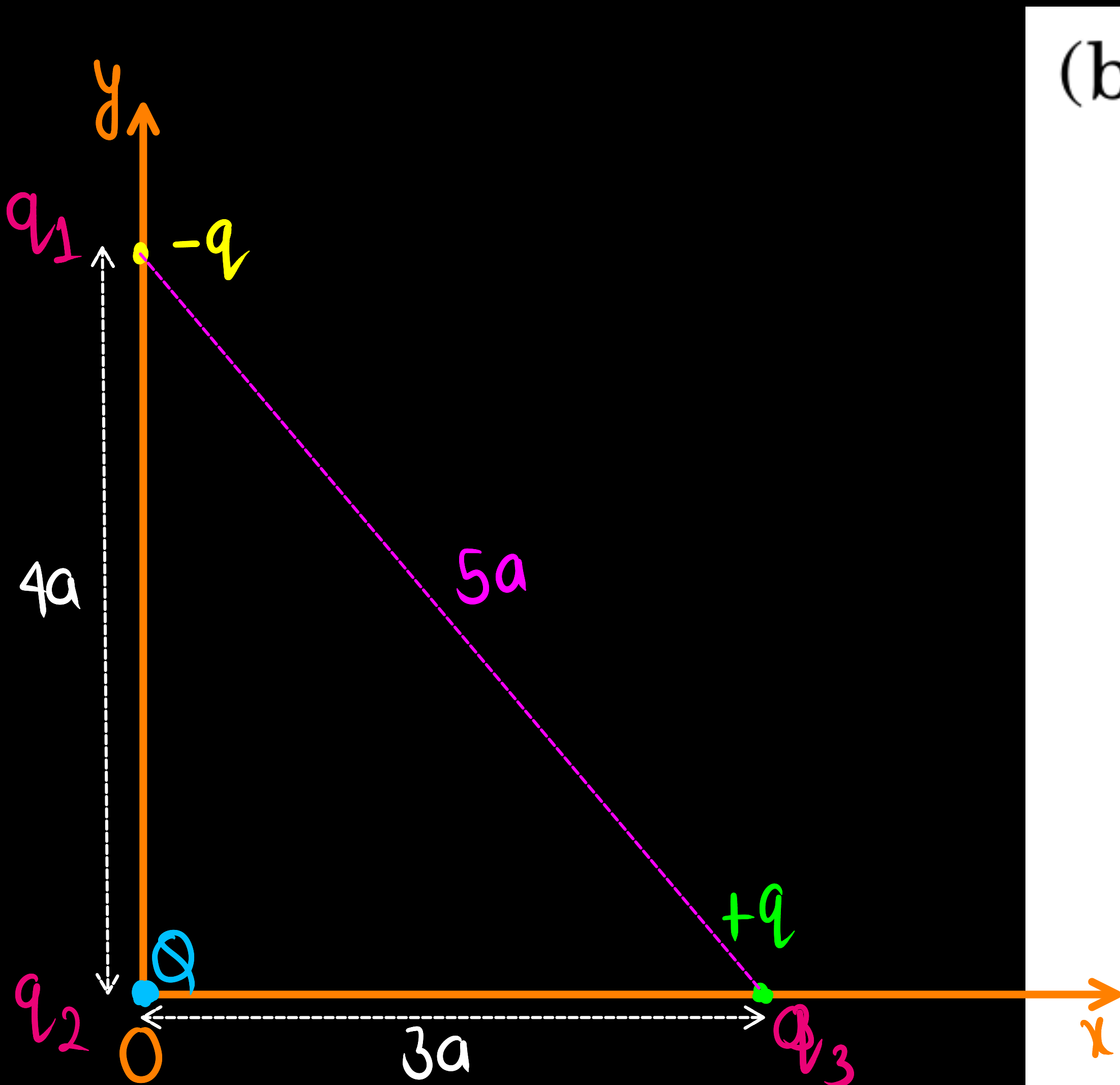
$$\hookrightarrow \frac{J}{C} \text{ or Volt}$$

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$$\frac{4qQ - 3qQ}{12a} = \frac{q^2}{5a}$$

$$\frac{qQ}{12a} = \frac{q^2}{5a}$$



Given: $U = 0$

$$\frac{1}{4\pi\epsilon_0} \left(\frac{q_1 q_2}{r_{12}} + \frac{q_1 q_3}{r_{13}} + \frac{q_2 q_3}{r_{23}} \right) = 0$$

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$$\frac{4qQ - 3qQ}{12a} = \frac{q^2}{5a}$$

$$\frac{qQ}{12a} = \frac{q^2}{5a}$$

$$Q = \frac{12}{5} q$$

(b) (i) Define electric potential at a point and write its SI unit.

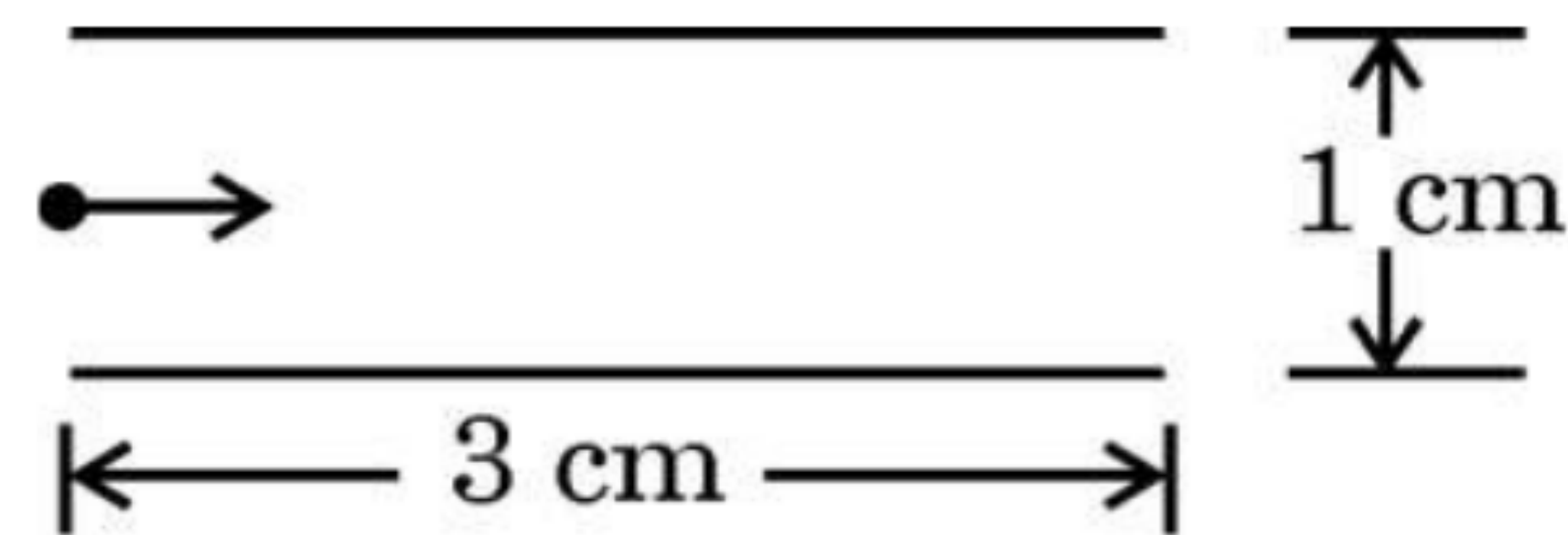
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34. A beam of electrons moving horizontally with a velocity of 3×10^7 m/s enters a region between two plates as shown in the figure. A suitable potential difference is applied across the plates such that the electron beam just strikes the edge of the lower plate.



Answer the following questions based on the above :

- | | | |
|-----|--|---|
| (a) | How long does an electron take to strike the edge ? | 1 |
| (b) | What is the shape of the path followed by the electron and why ? | 1 |
| (c) | Find the potential difference applied. | 2 |

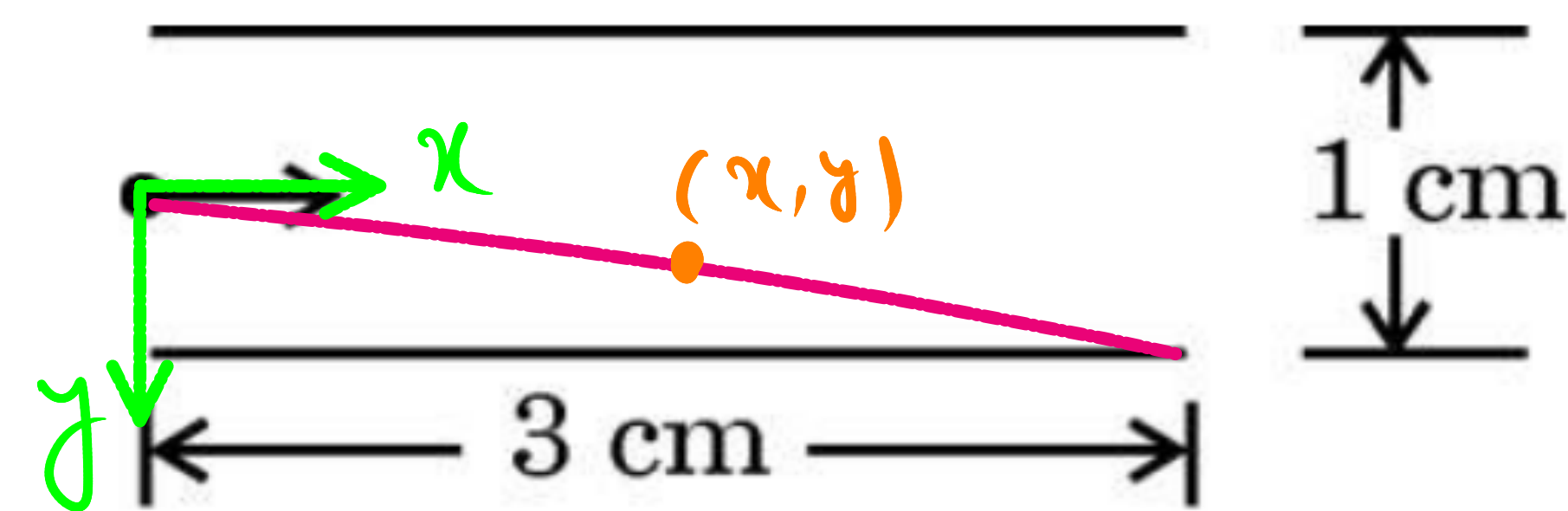
a) Motion along x-axis -

$$u_x = 3 \times 10^7 \text{ ms}^{-1}$$

$$x_0 = 0$$

$$a_x = 0 \quad (\because F_x = 0)$$

34. A beam of electrons moving horizontally with a velocity of $3 \times 10^7 \text{ m/s}$ enters a region between two plates as shown in the figure. A suitable potential difference is applied across the plates such that the electron beam just strikes the edge of the lower plate.



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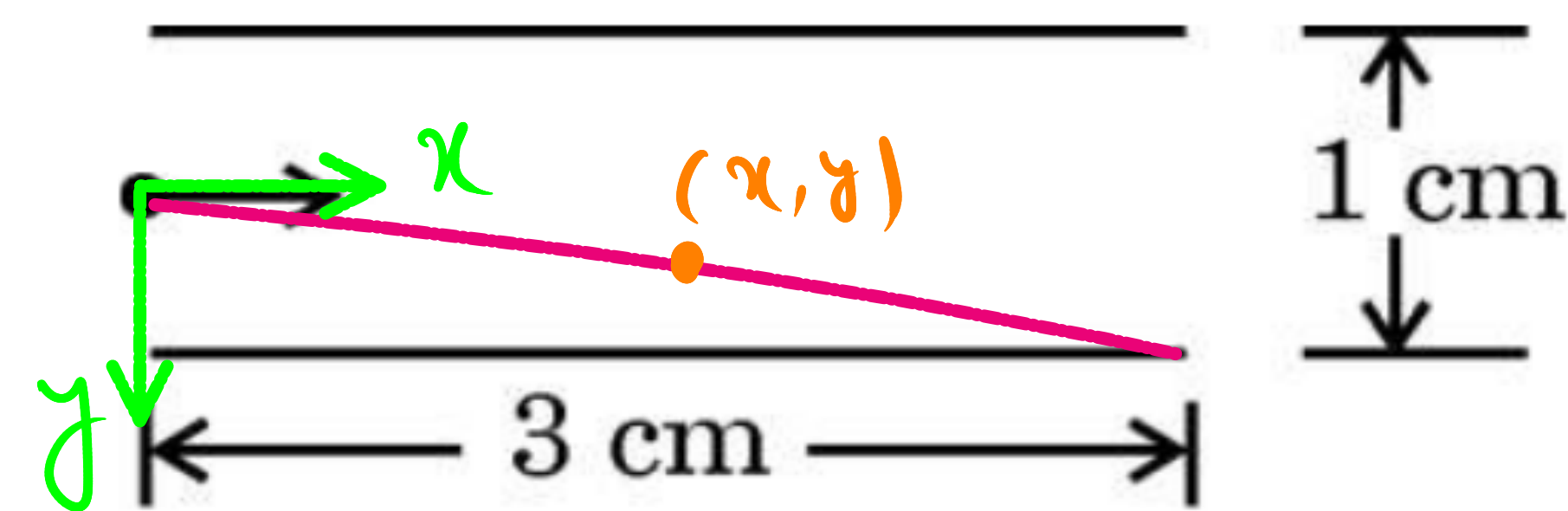
$$x_0 = 0$$

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using $s_x = u_x t + \frac{1}{2} a_x t^2$

$$x - x_0 = u_x t + \frac{1}{2} \times 0 \times t^2$$

34. A beam of electrons moving horizontally with a velocity of $3 \times 10^7 \text{ m/s}$ enters a region between two plates as shown in the figure. A suitable potential difference is applied across the plates such that the electron beam just strikes the edge of the lower plate.



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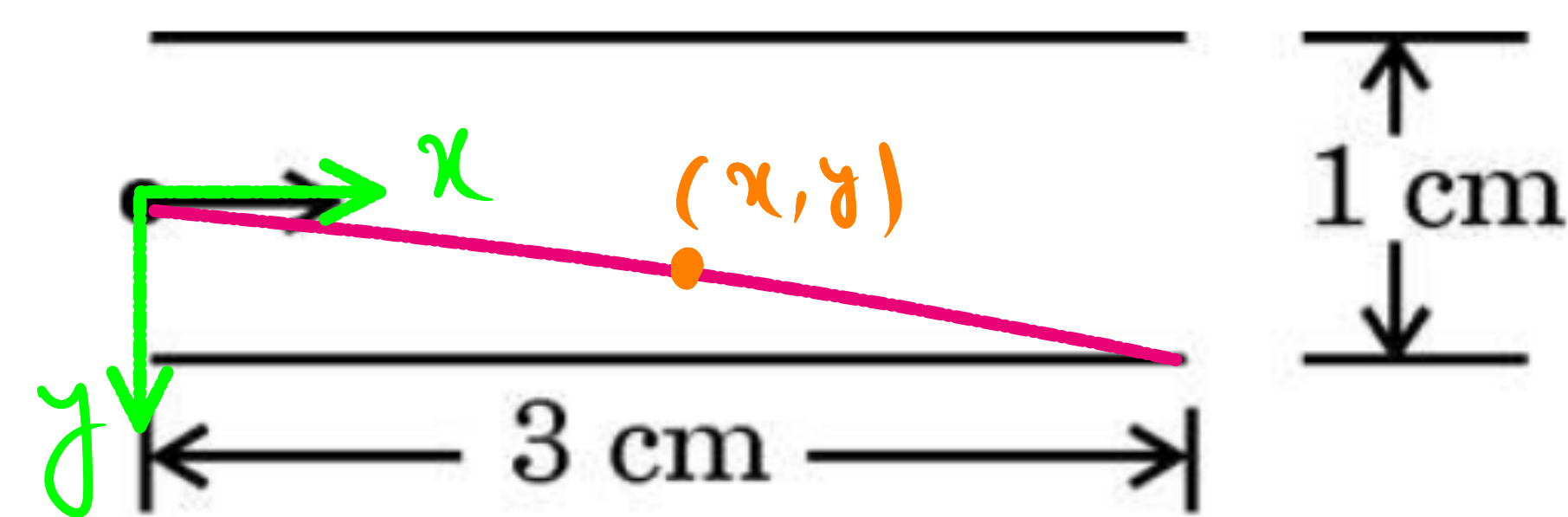
Using $s_x = u_x t + \frac{1}{2} a_x t^2$

$$x - x_0 = u_x t + \frac{1}{2} \times 0 \times t^2$$

$$x = u_x t$$

$$t = \frac{x}{u_x} \quad \text{--- (i)}$$

34. A beam of electrons moving horizontally with a velocity of $3 \times 10^7 \text{ m/s}$ enters a region between two plates as shown in the figure. A suitable potential difference is applied across the plates such that the electron beam just strikes the edge of the lower plate.



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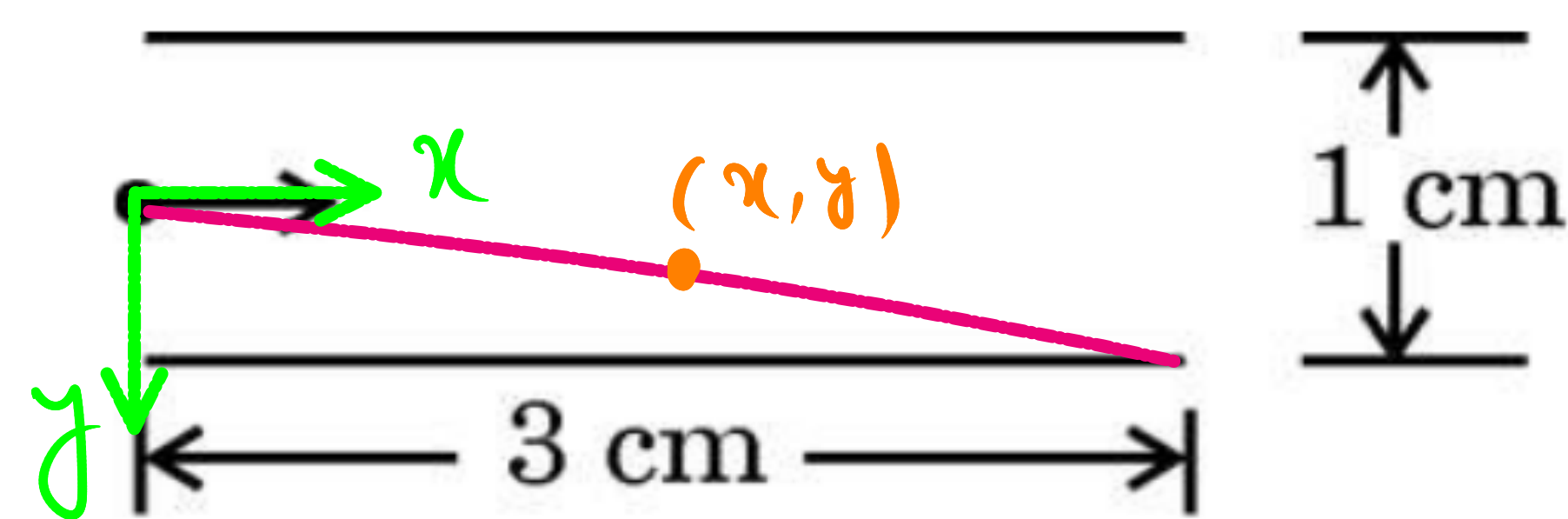
$$x = u_x t$$

$$t = \frac{x}{u_x} \quad \text{--- (i)}$$

$$t = \frac{3 \times 10^{-2} \text{ m}}{3 \times 10^7 \text{ ms}^{-1}}$$

$$t = 10^{-9} \text{ s}$$

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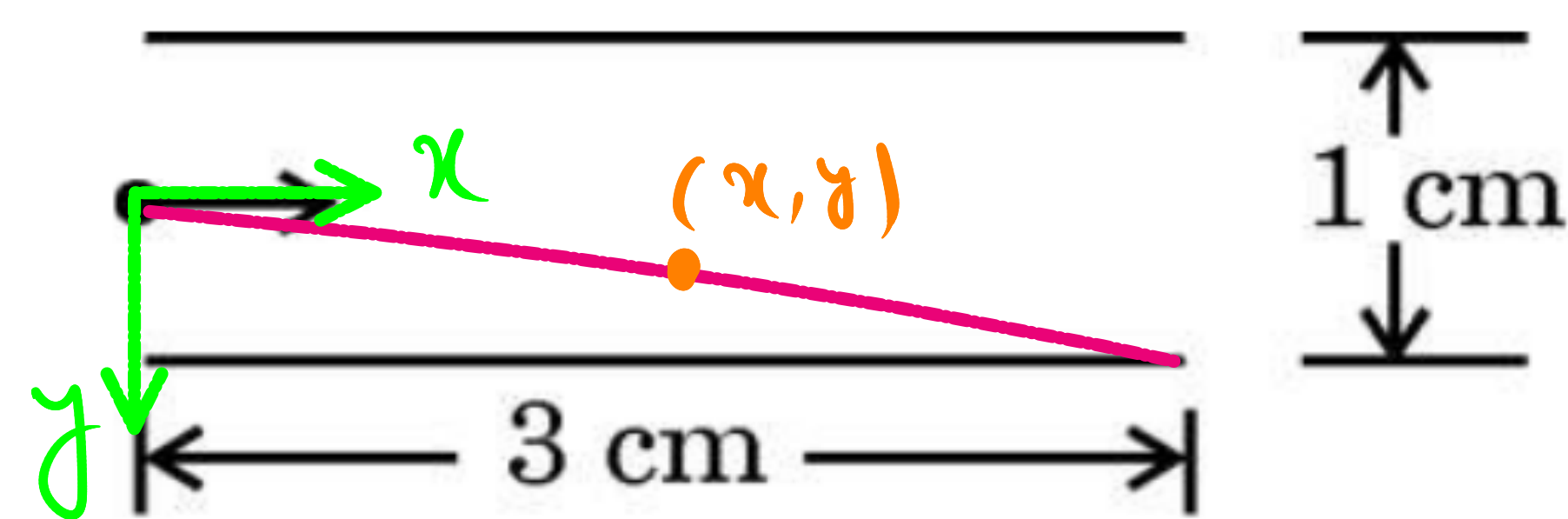
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b) Shape of the path - Parabola

$$y = \frac{1}{2} a_y t^2 \quad (\because u_y = 0)$$

$$y = \frac{1}{2} \frac{eE}{m_e} t^2 \quad (\because a_y = \frac{eE}{m_e})$$

$$y = \frac{1}{2} \frac{eE}{m_e} \frac{x^2}{u_x^2} \quad \text{--- (2)}$$

↳ eqⁿ of parabola

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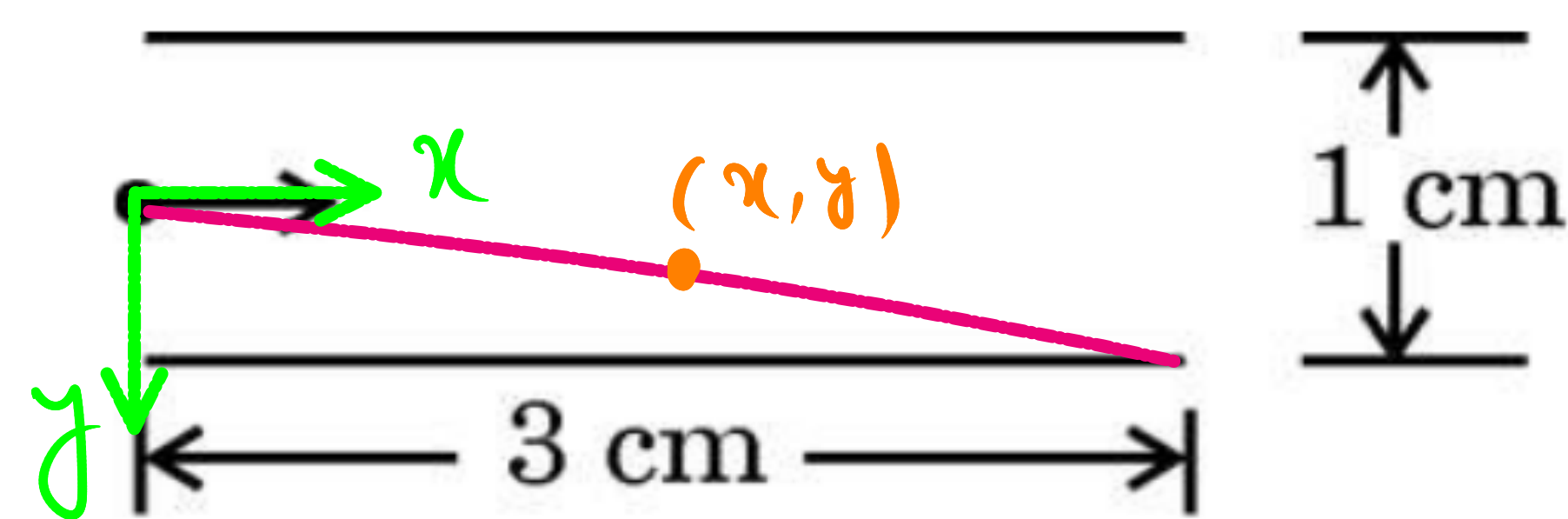
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↳ eqⁿ of parabola

When $x = 3 \text{ cm} = 3 \times 10^{-2} \text{ m}$, $y = 0.5 \text{ cm} = 5 \times 10^{-3} \text{ m}$
from eqⁿ (2)

$$5 \times 10^{-3} = \frac{1}{2} \times \frac{1.6 \times 10^{-19} \times E}{9.1 \times 10^{-31}} \times \frac{9 \times 10^{-4}}{9 \times 10^{14}}$$

a) Motion along x-axis -

$$u_x = 3 \times 10^7 \text{ ms}^{-1}$$

$$x_0 = 0$$

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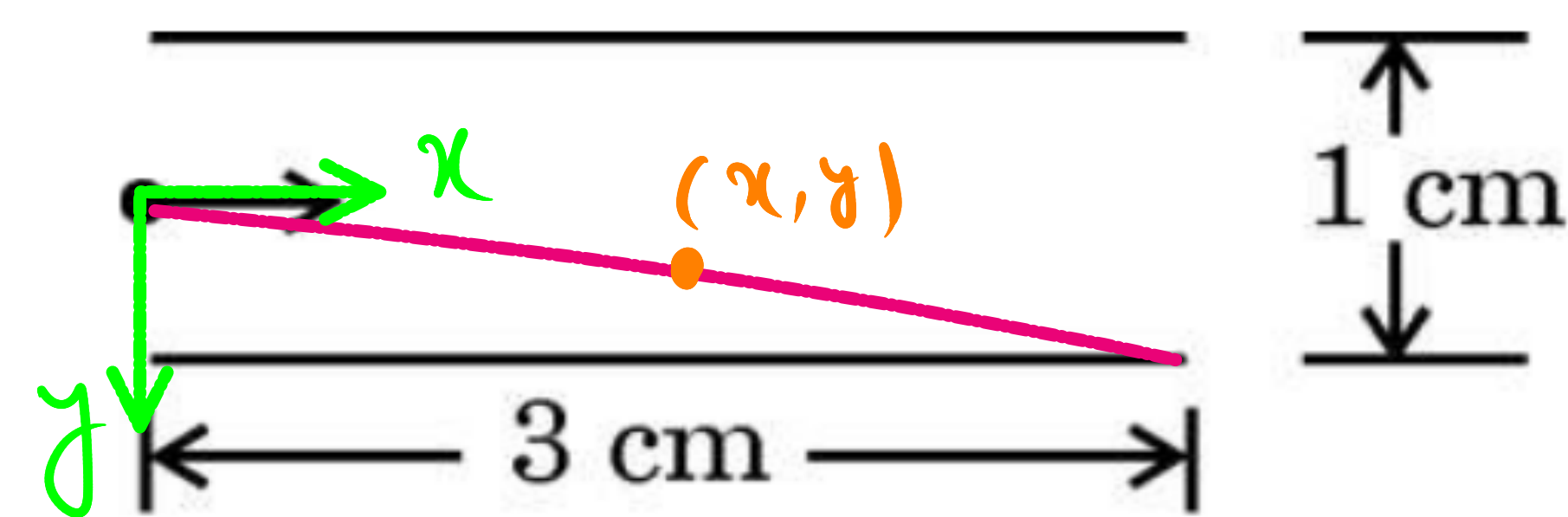
$$x = u_x t$$

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$$5 \times 10^{-3} = \frac{1}{2} \times \frac{0.8 \times 10^{-19} \times E}{9.1 \times 10^{-31}} \times \frac{9 \times 10^{-4}}{9 \times 10^{14}}$$

$$5 \times 10^{-3} = \frac{0.8}{9.1} \times 10^{-6} \times E$$

$$E = \frac{5 \times 9.1}{0.8} \times 10^3$$

$$E = \underline{\underline{56.875 \times 10^3 \frac{\text{V}}{\text{m}}}}$$

a) Motion along x-axis -

$$u_x = 3 \times 10^7 \text{ ms}^{-1}$$

$$x_0 = 0$$

$$a_x = 0 \quad (\because F_x = 0)$$

using $s_x = u_x t + \frac{1}{2} a_x t^2$

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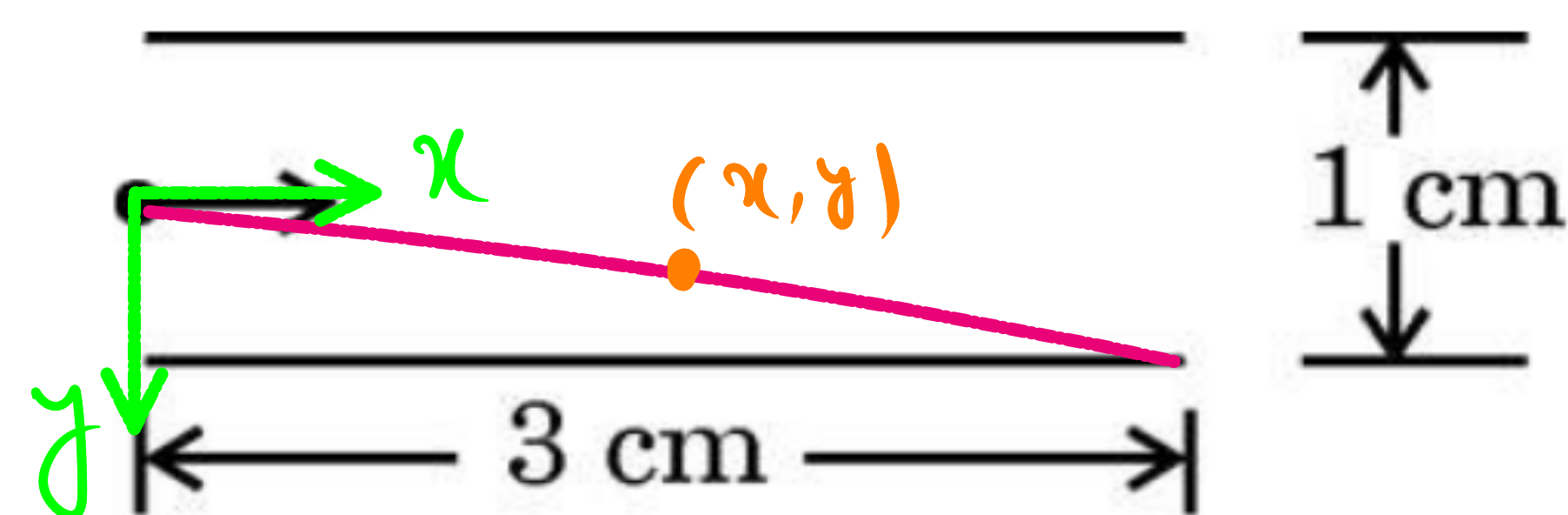
$$x = u_x t$$

$$t = \frac{x}{u_x} \quad \text{--- (1)}$$

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Answer the following questions based on the above :

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$$5 \times 10^{-3} = \frac{0.8}{9.1} \times 10^{-6} \times E$$

$$E = \frac{5 \times 9.1 \times 10^3}{0.8}$$

$$E = 56.875 \times 10^3 \frac{\text{V}}{\text{m}}$$

$$V = Ed$$

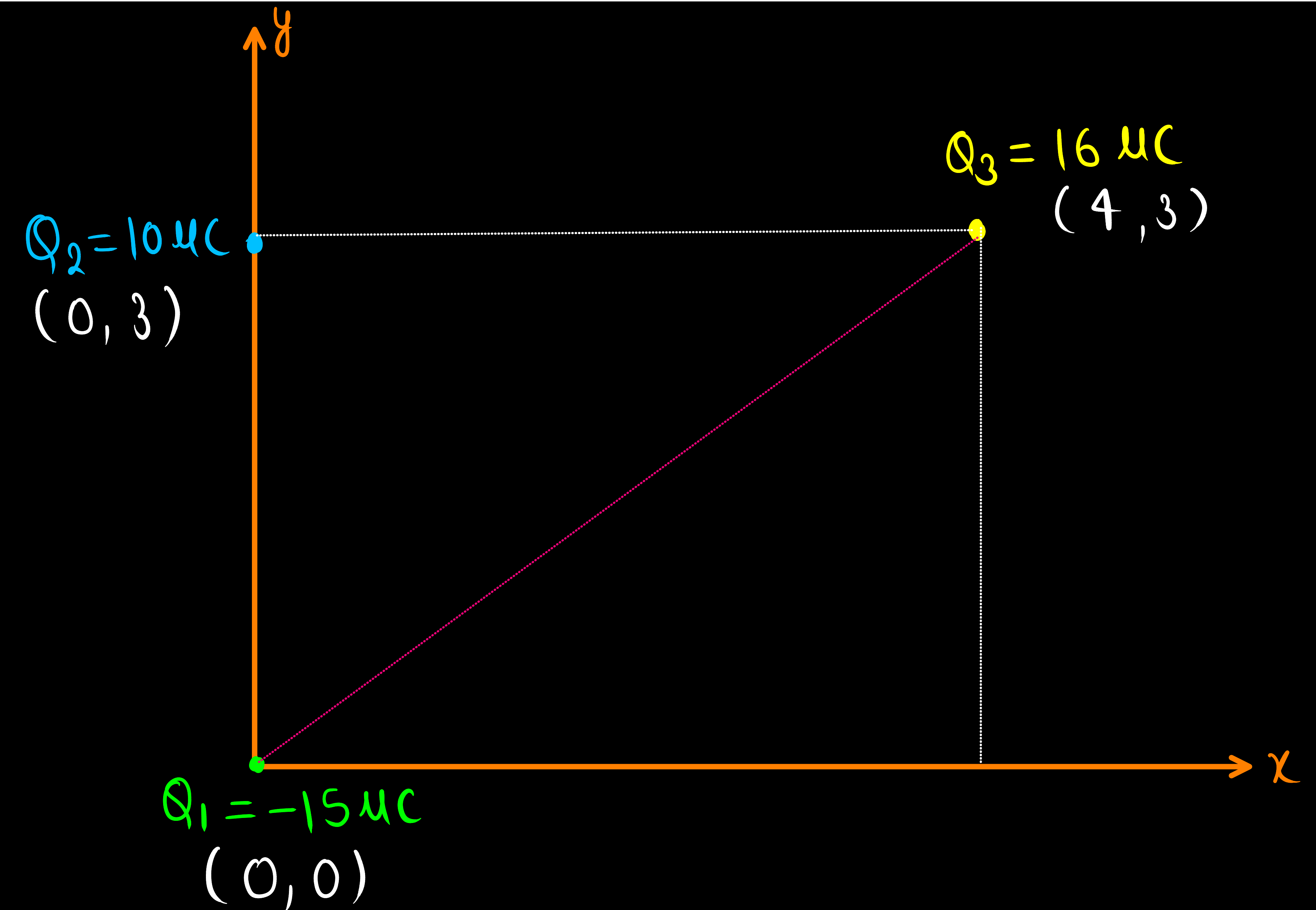
$$V = 56.875 \times 10^3 \frac{\text{V}}{\text{m}} \times 10^{-2} \text{ m}$$

$$V = 568.75 \text{ V}$$

28. Three point charges Q_1 ($-15\ \mu\text{C}$), Q_2 ($10\ \mu\text{C}$) and Q_3 ($16\ \mu\text{C}$) are located at (0 cm, 0 cm), (0 cm, 3 cm) and (4 cm, 3 cm) respectively. Calculate the electrostatic potential energy of this system of charges.

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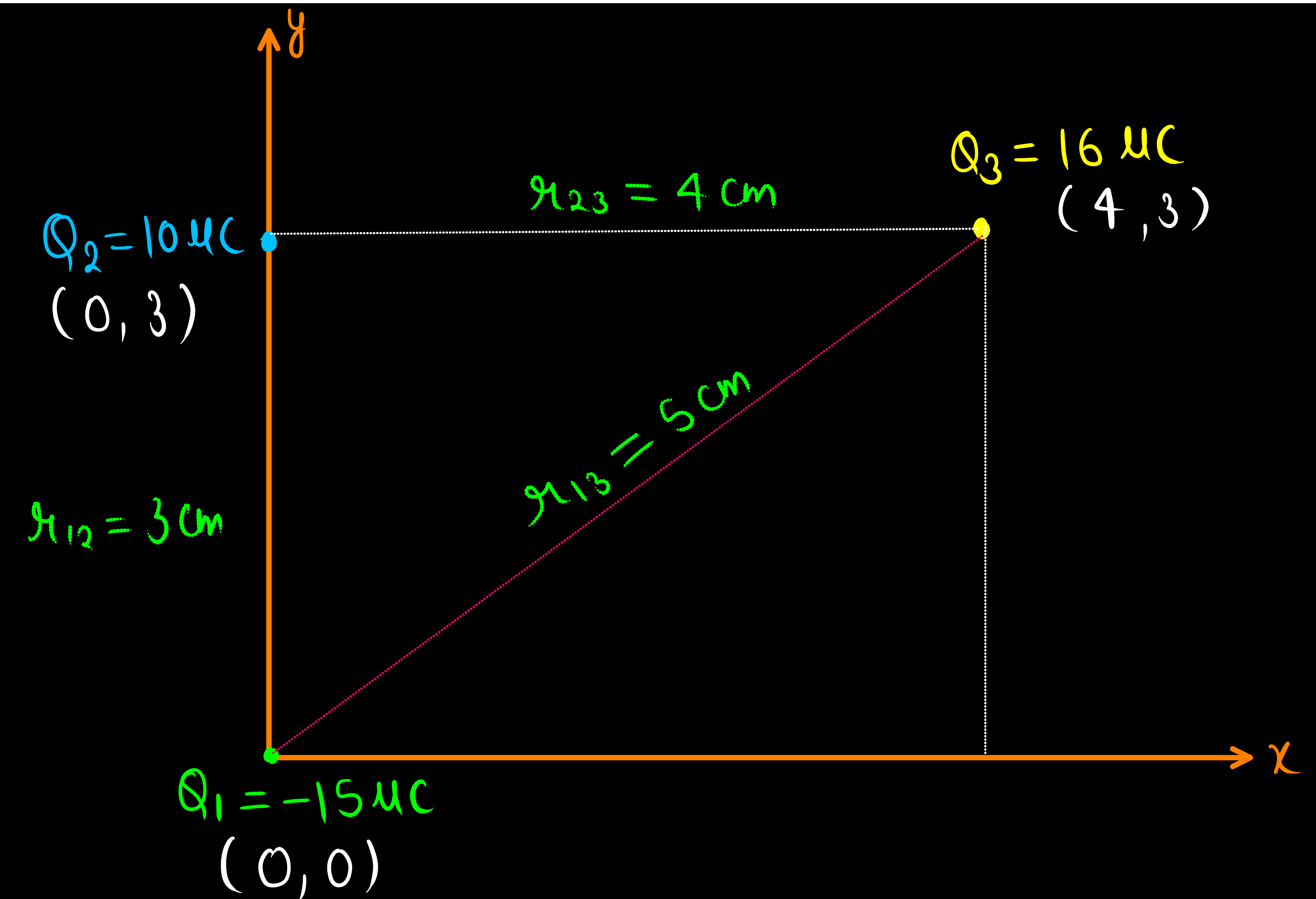
3



28. Three point charges Q_1 ($-15 \mu\text{C}$), Q_2 ($10 \mu\text{C}$) and Q_3 ($16 \mu\text{C}$) are located at (0 cm, 0 cm), (0 cm, 3 cm) and (4 cm, 3 cm) respectively. Calculate the electrostatic potential energy of this system of charges.

3

$$U = \frac{1}{4\pi\epsilon_0} \left[\frac{Q_1 Q_2}{r_{12}} + \frac{Q_1 Q_3}{r_{13}} + \frac{Q_2 Q_3}{r_{23}} \right] + \frac{10 \times 10^{-6} \times 16 \times 10^{-6}}{4 \times 10^{-2}} \Bigg]$$



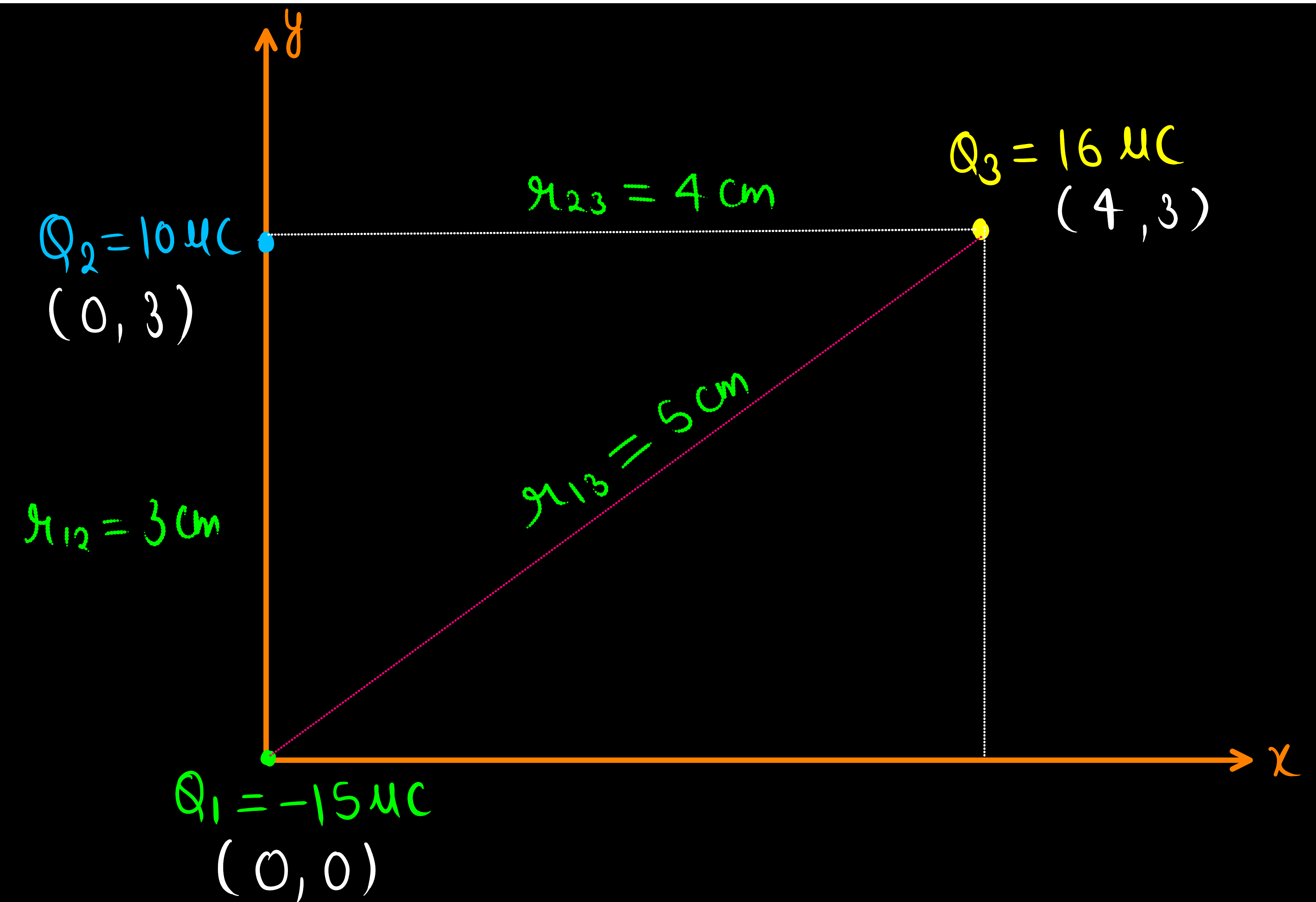
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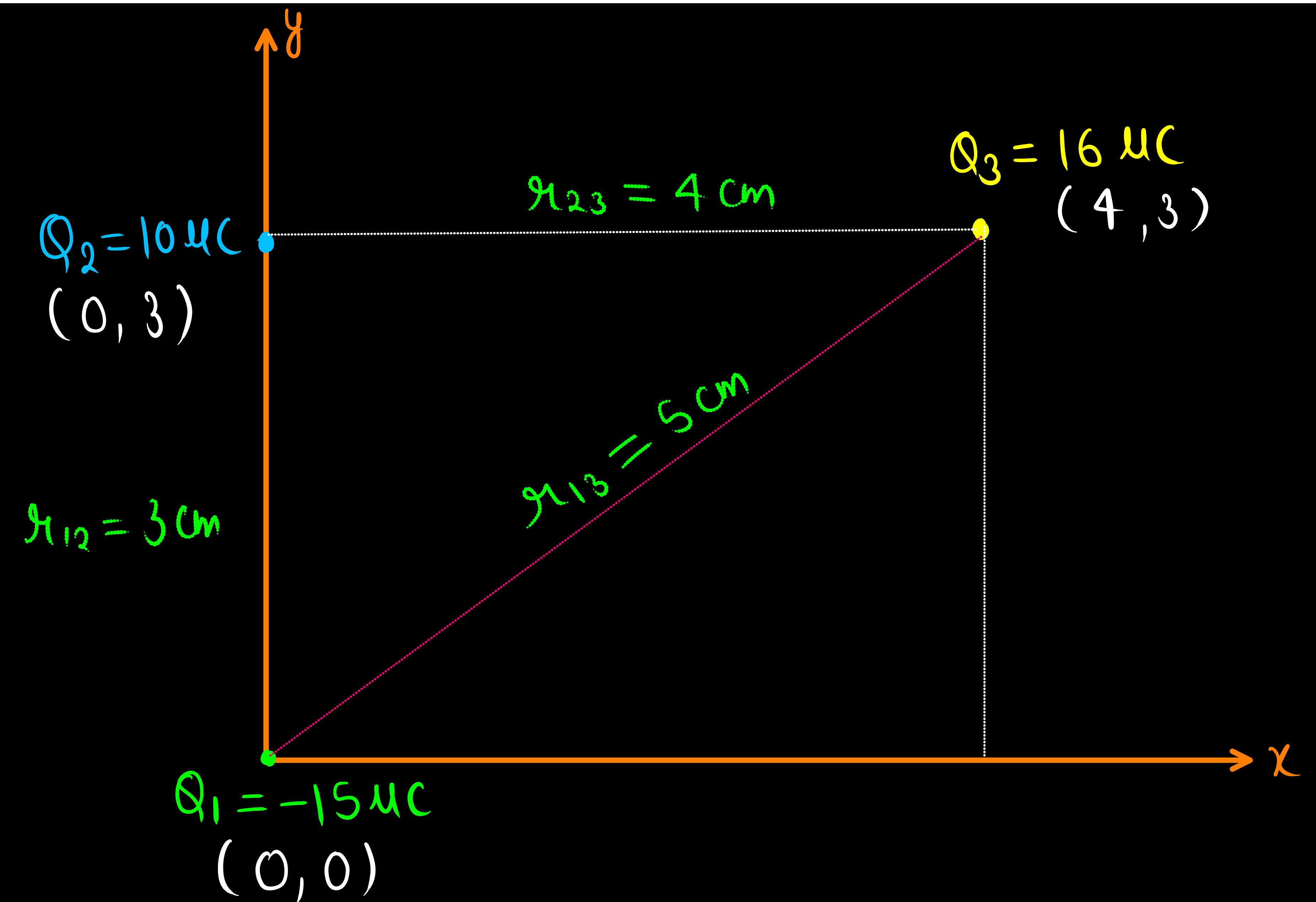
$\frac{5}{\quad}$ $\frac{3}{\quad}$

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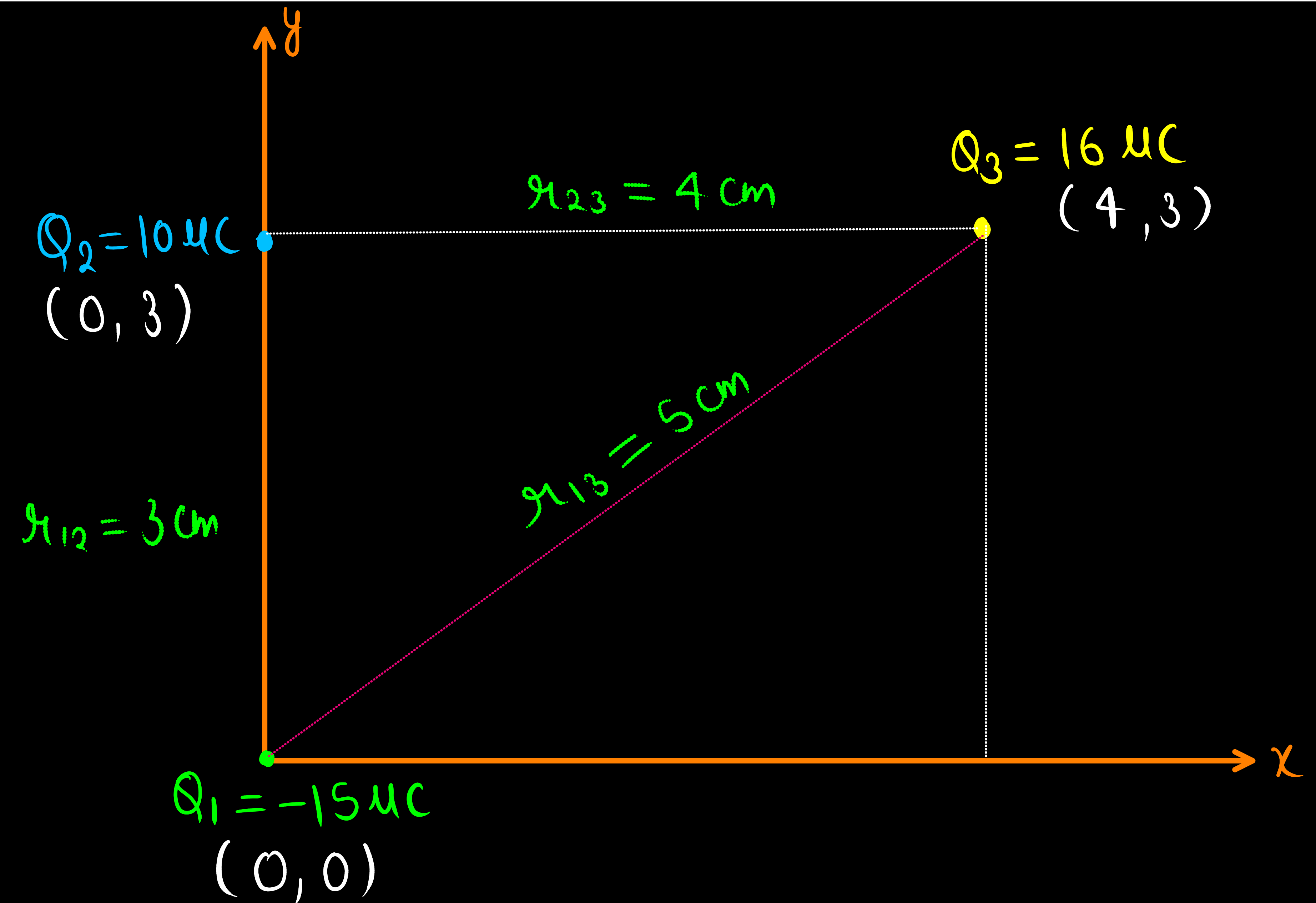
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30. A $100\ \mu\text{F}$ capacitor is charged by a $12\ \text{V}$ battery.

- (a) How much electrostatic energy is stored by the capacitor ?
- (b) The capacitor is disconnected from the battery and connected in parallel to another uncharged $100\ \mu\text{F}$ capacitor. What is the electrostatic energy stored by the system ?

a) $C = 100 \mu\text{F}$
 $V = 12 \text{ V}$
 $U = \frac{1}{2} C V^2$
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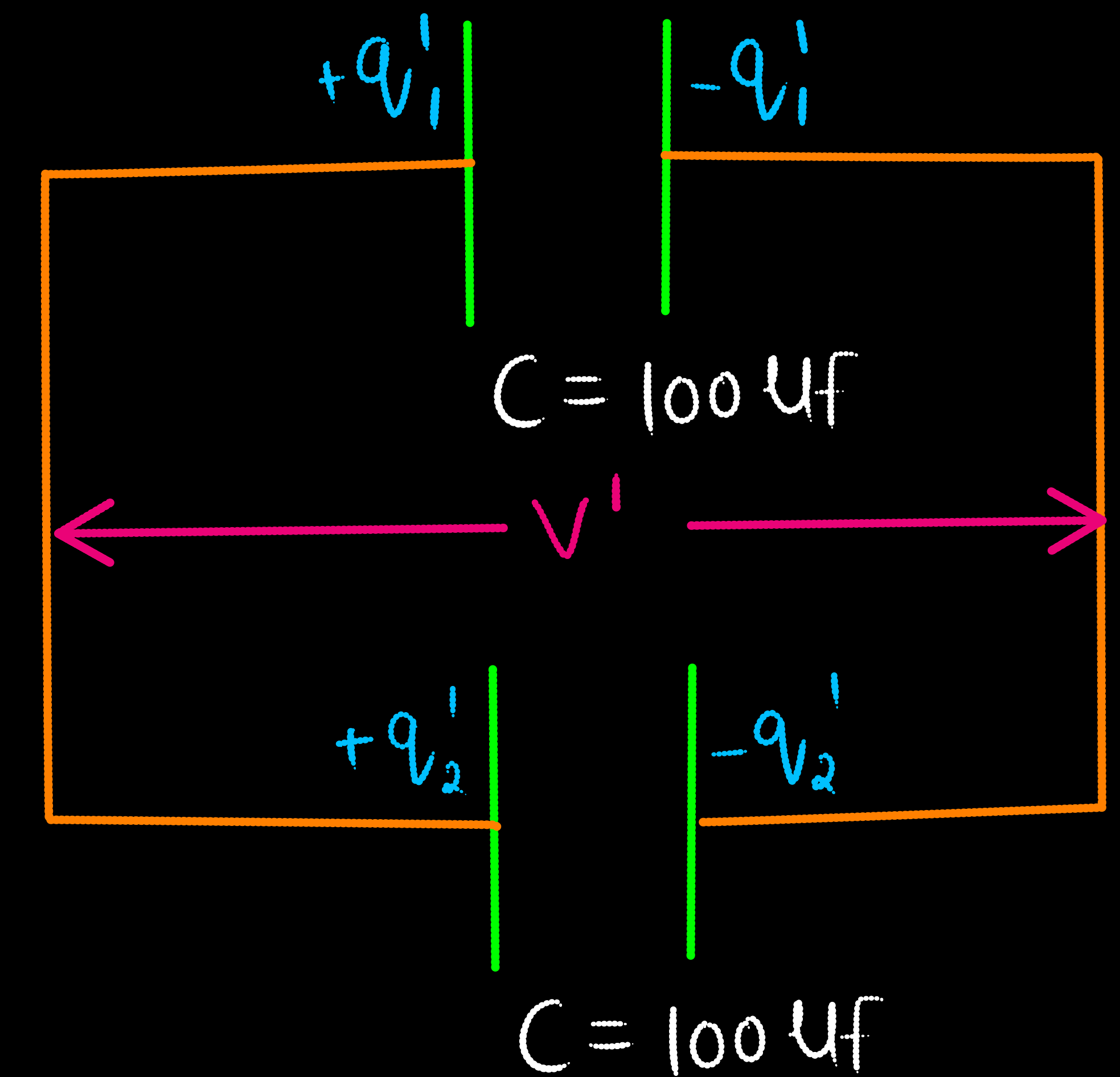
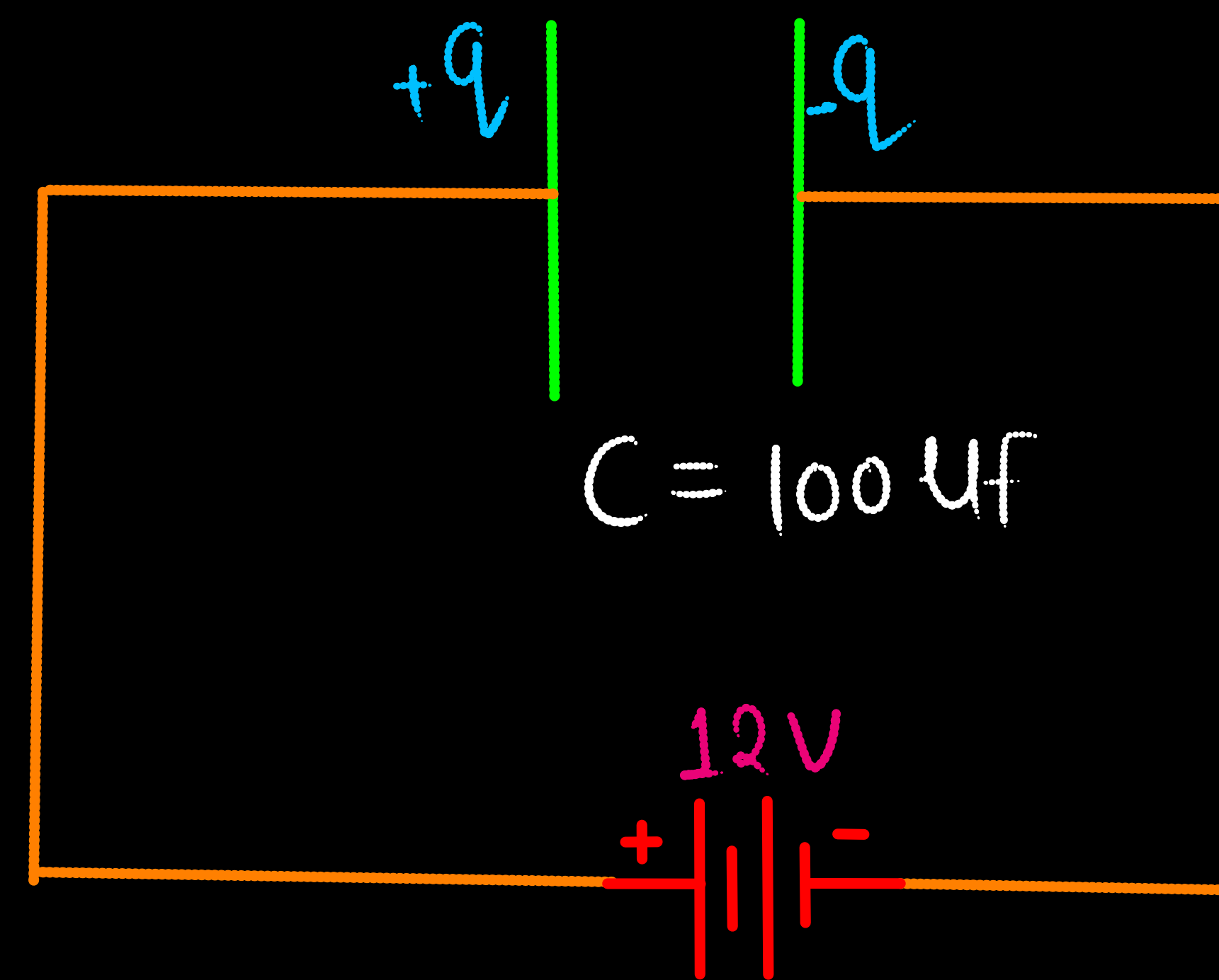
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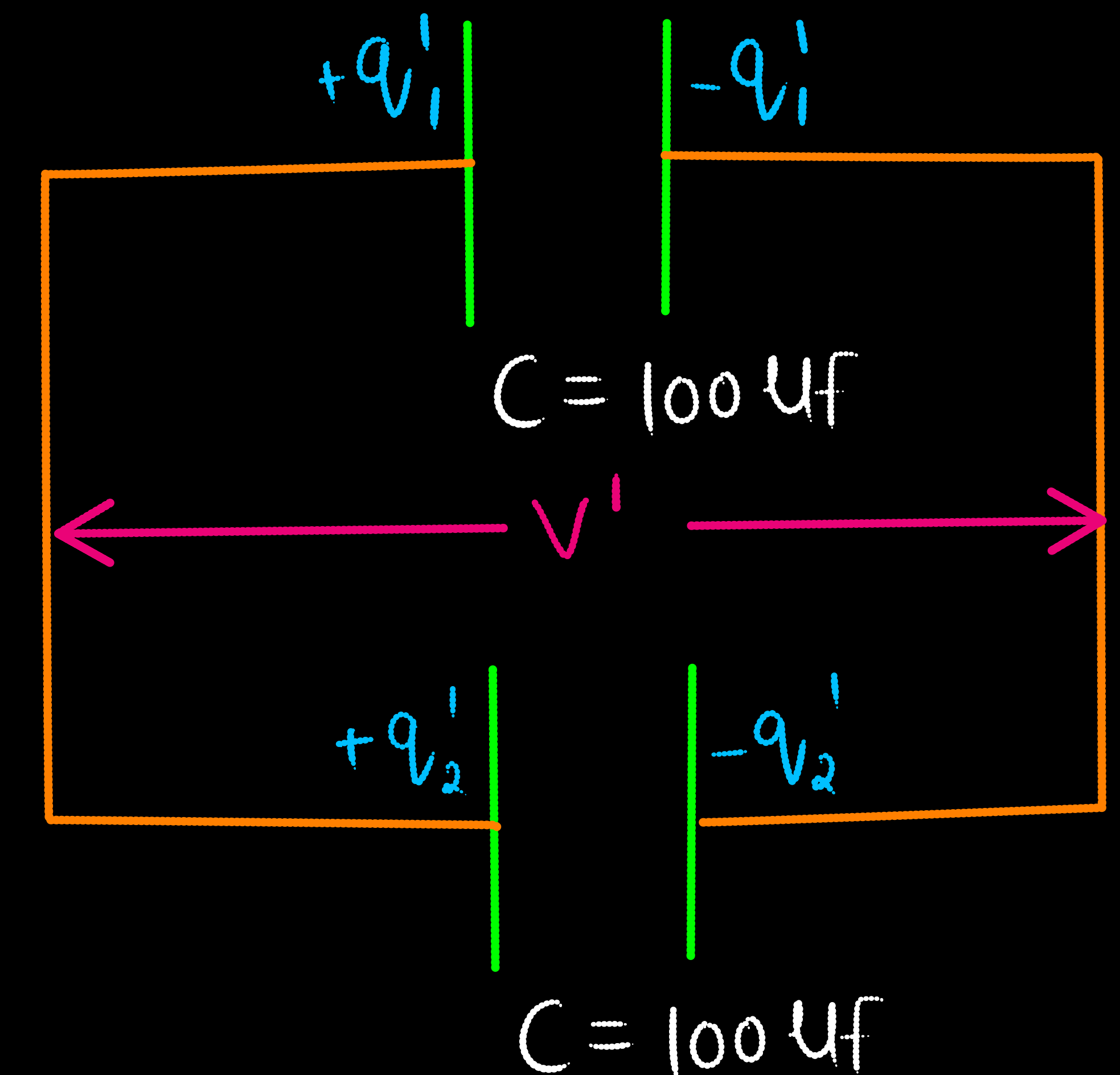
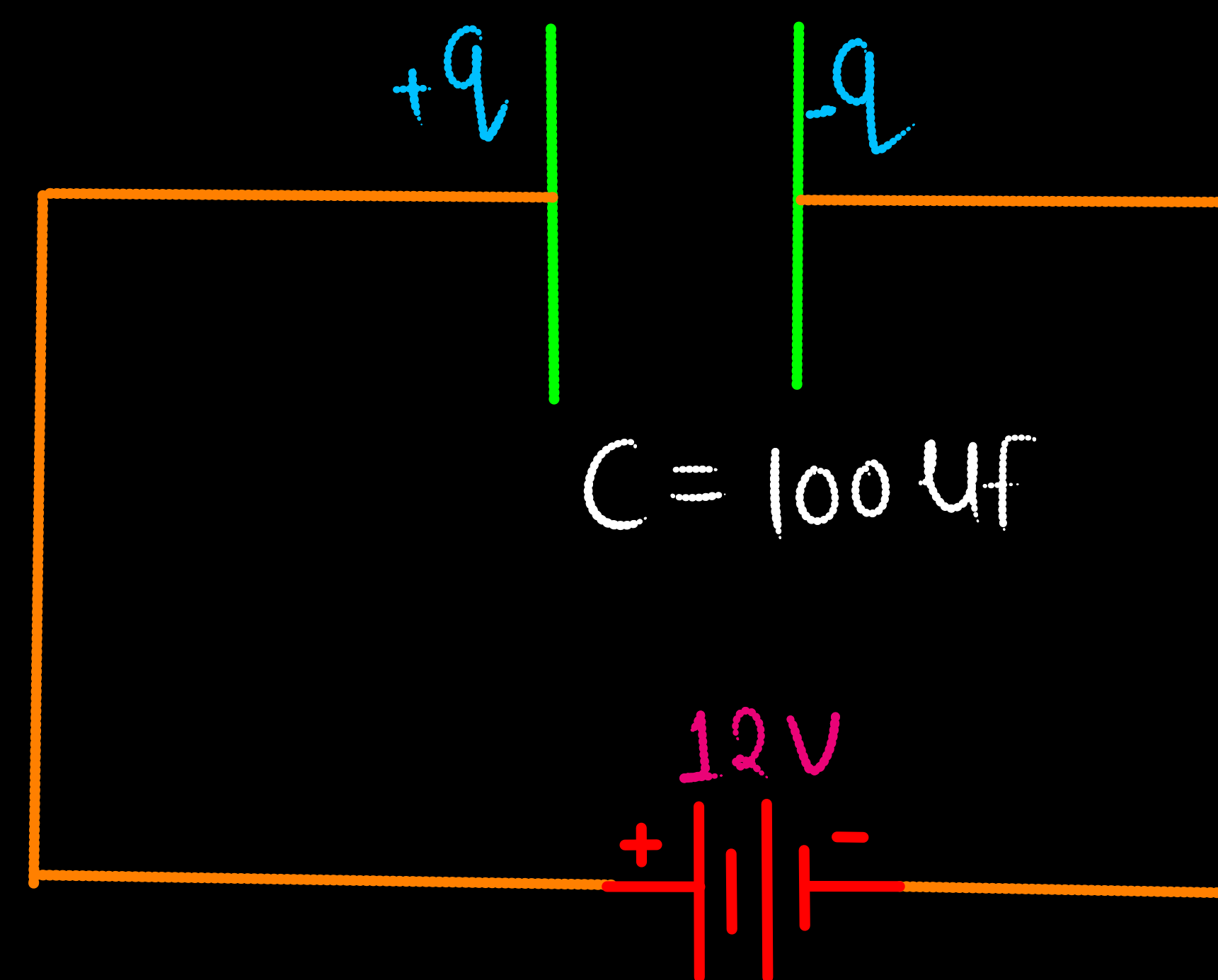
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b) $q_1' + q_2' = q$
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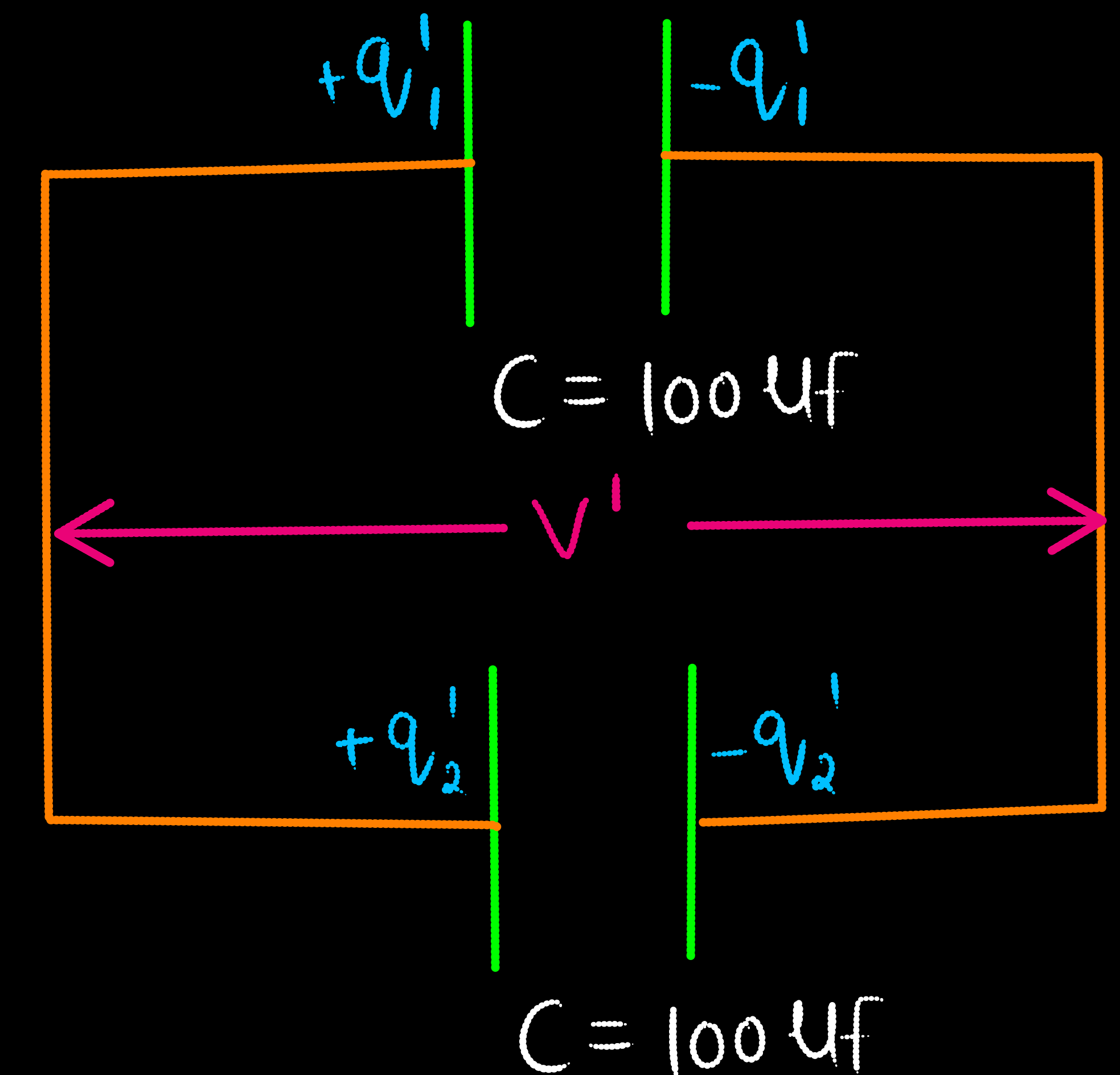
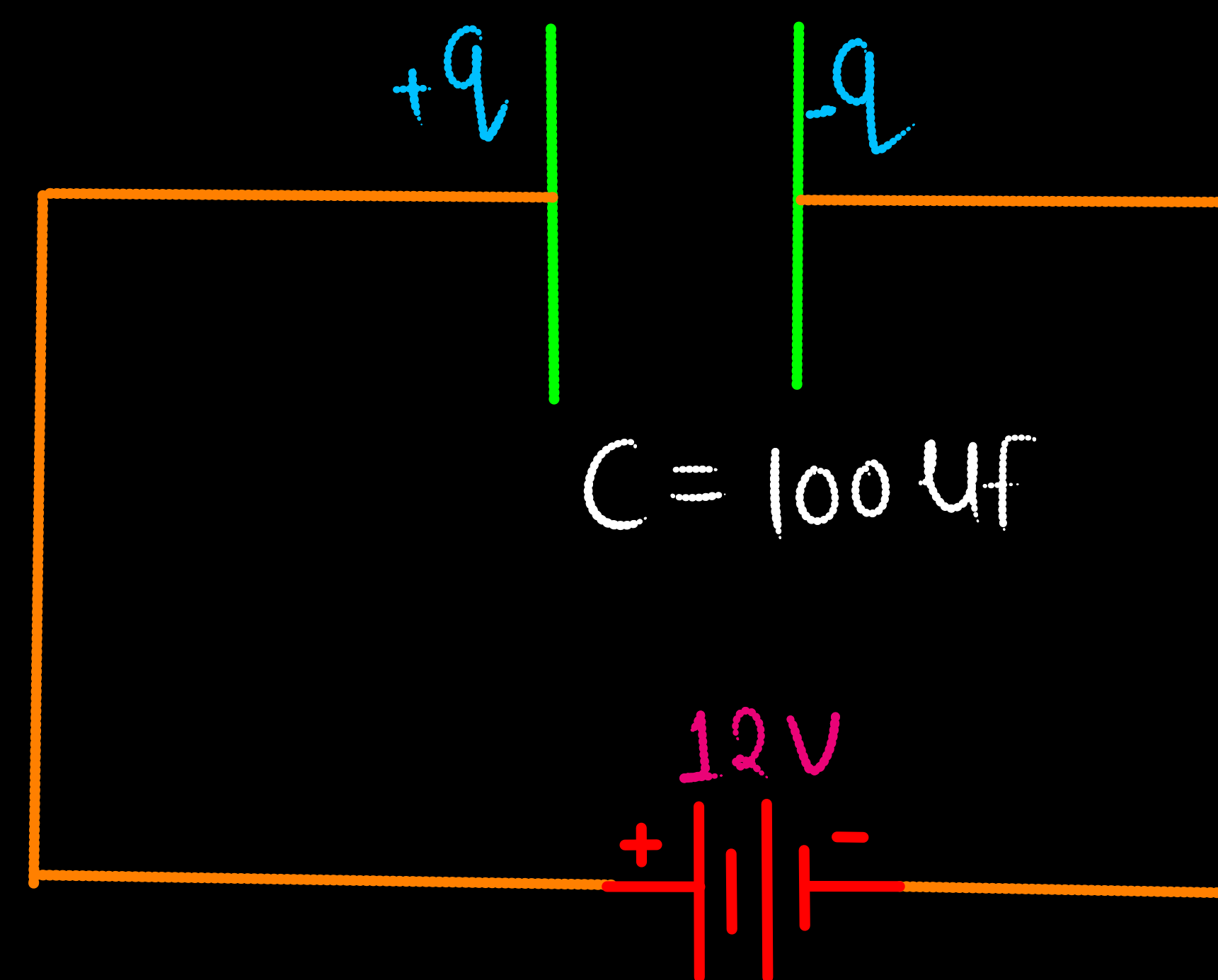
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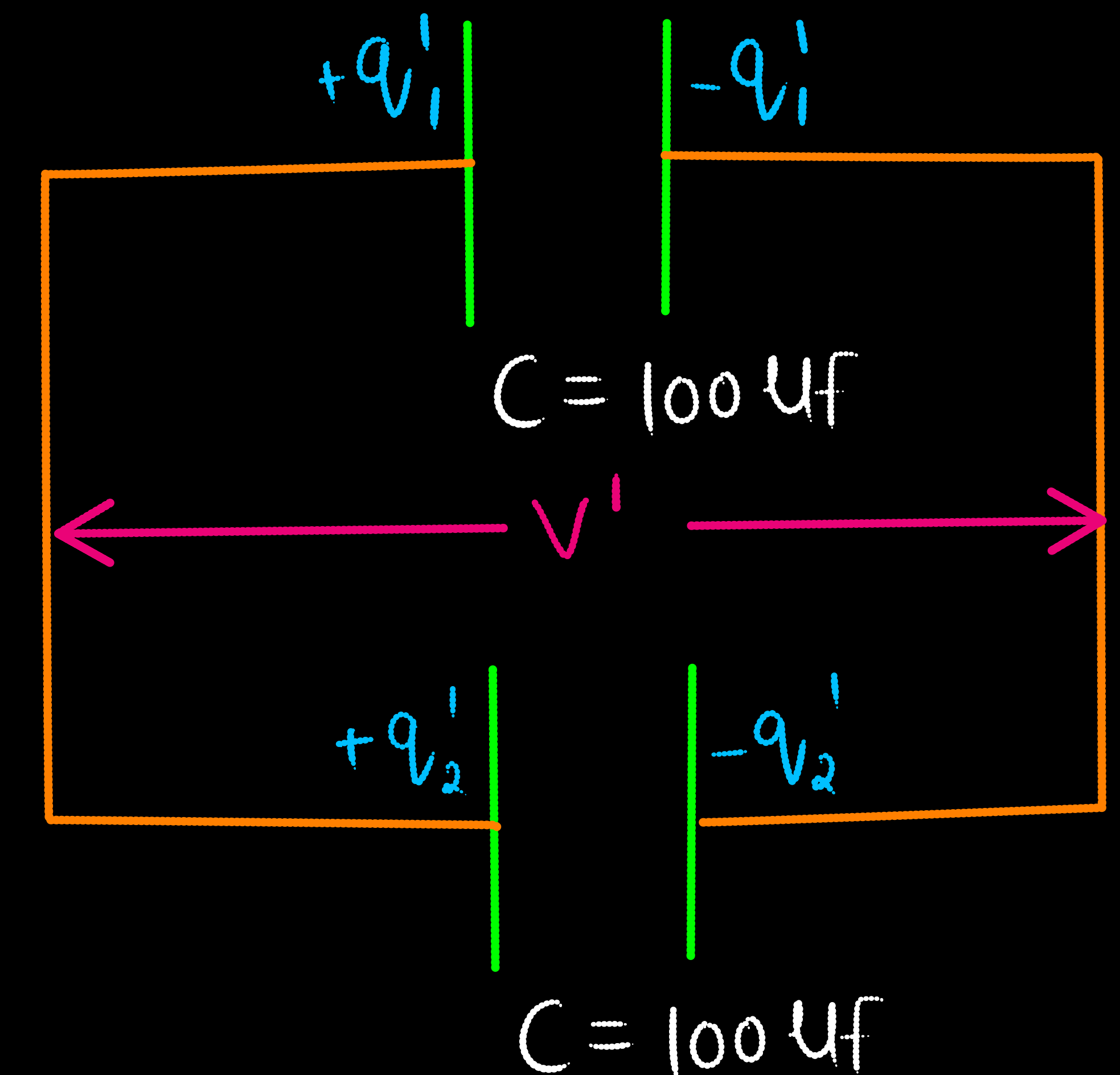
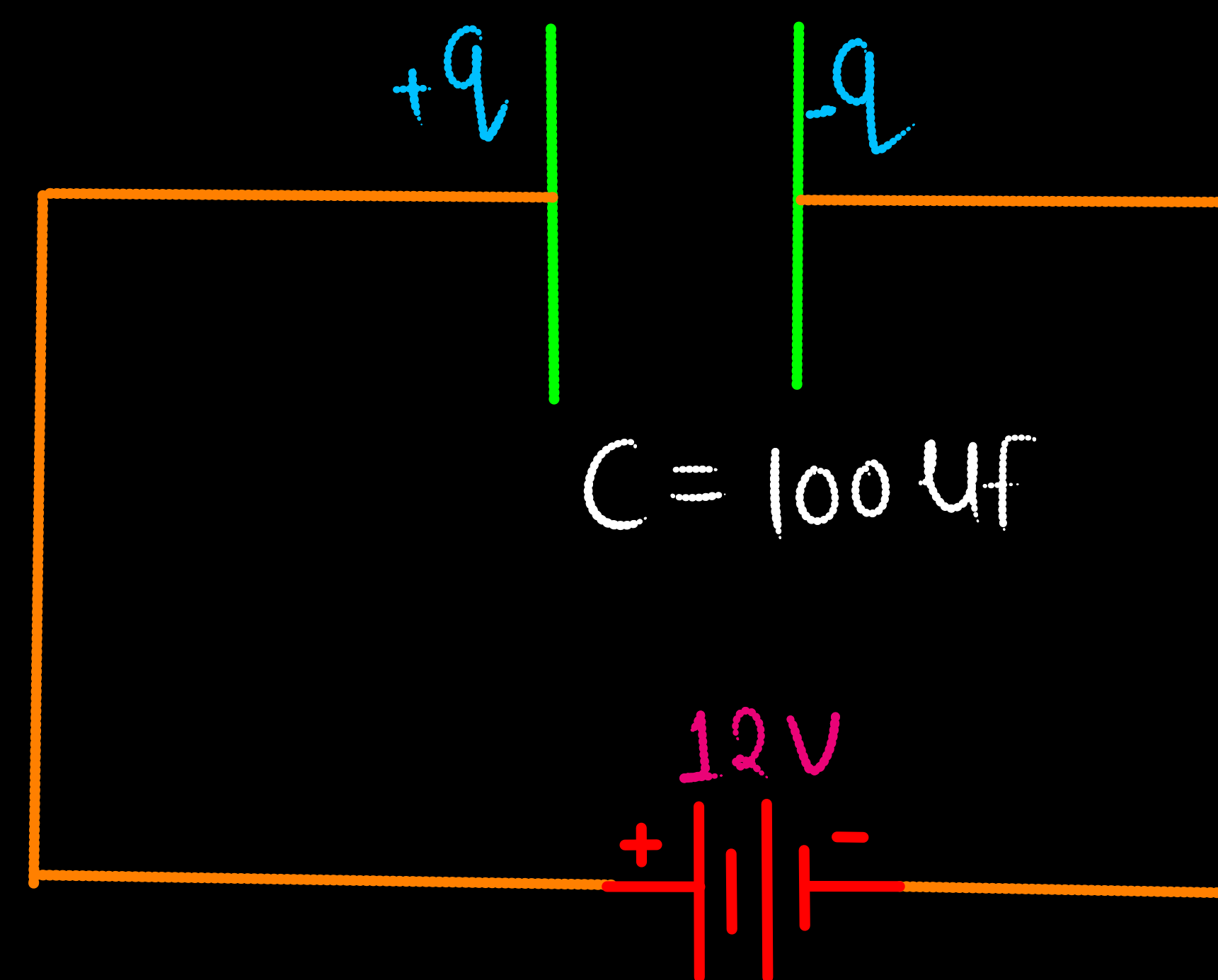
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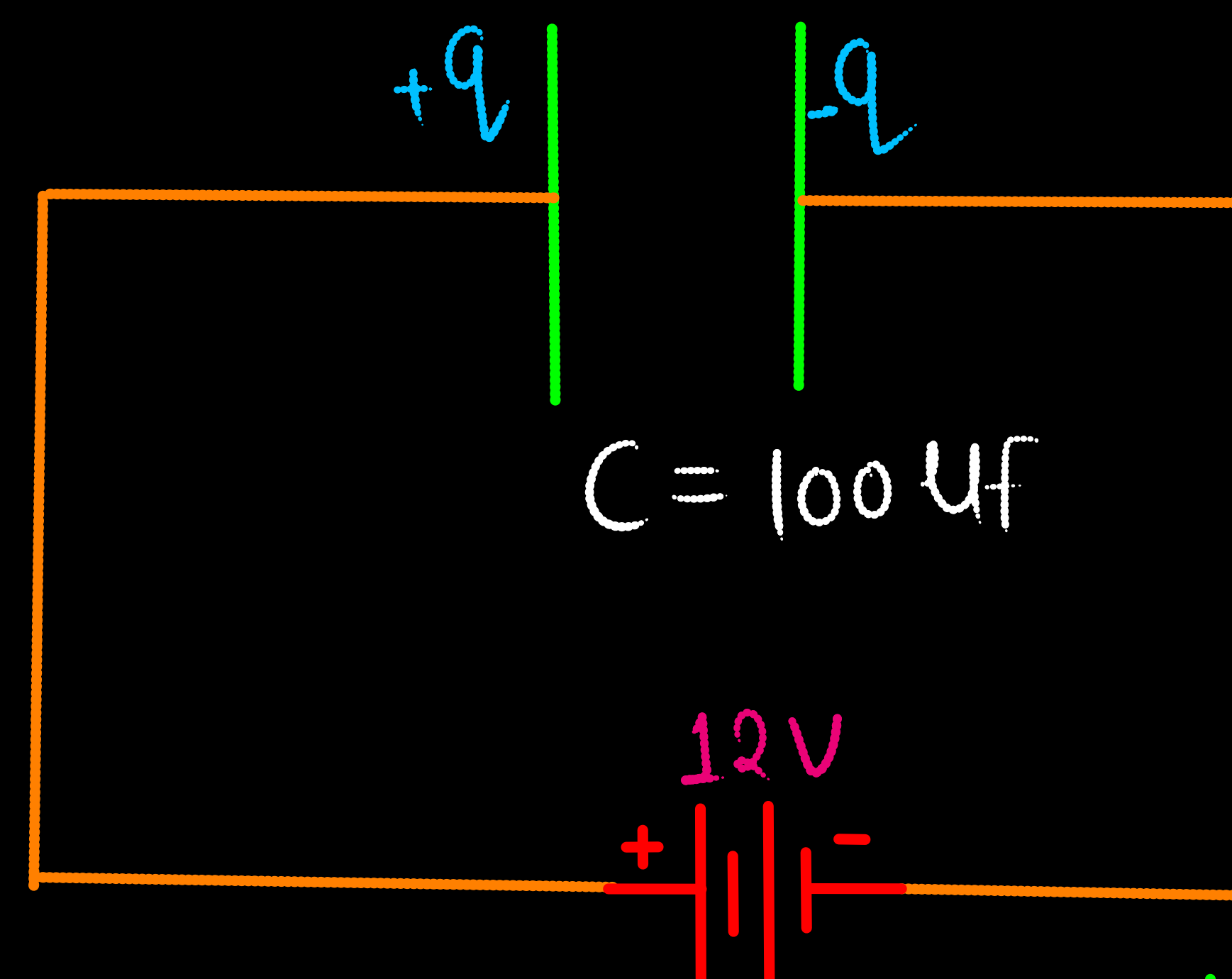
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 $C' = C + C = 2C$

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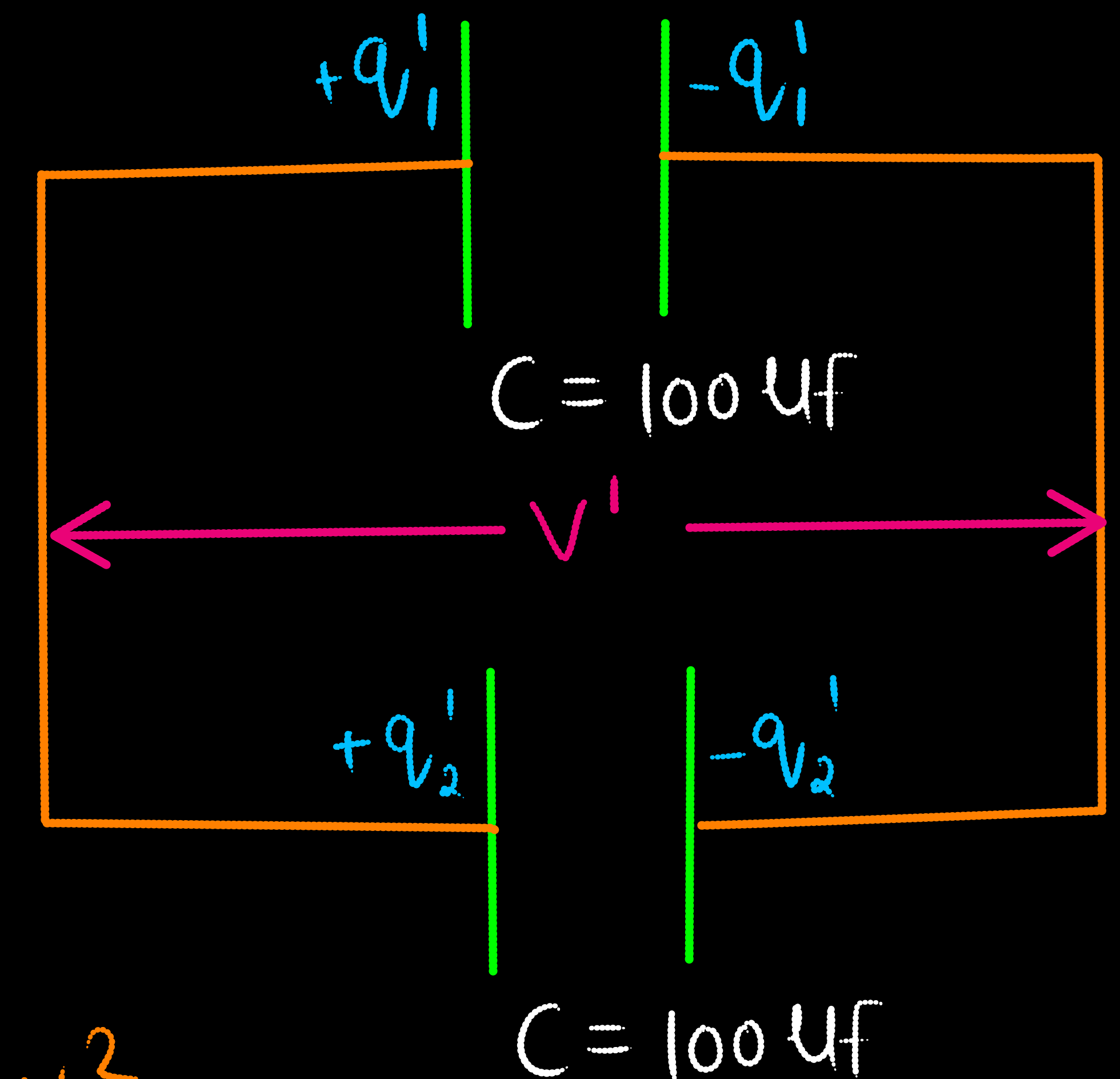
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$$U' = \frac{1}{2} C' V'^2$$

$$U' = \frac{1}{2} \times 2C \times \frac{V^2}{4}$$



$$U' = \frac{1}{2} \times \frac{1}{2} C V^2$$

$$U' = \frac{1}{2} U = 36 \times 10^{-4} \text{ J}$$